

Date Fruit Variety & Quality Models: Evaluation Report

Generated 2026-10-01 from the metric files in `outputs/` · models: probe_dinov3_b_bgaug+effv2b0_260_v2, grade_mnv3+grade_effv2b0, maturity_probe_dinov3_b+maturity_effv2b0

Summary of results (held-out test sets)

Component	Model	Test n	Accuracy	Macro-F1	Loss	Accuracy with manual review of low-confidence
Phase 1 · Variety classification	Ensemble: DINOv3 ViT-B/16 336 probe (background-augmented) + EfficientNetV2-B0 260	411	99.5%	0.994	0.102	99.5% on 100% auto-accepted
Phase 2 · Quality grade (1–3)	Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260	451	84.5%	0.813	0.371	97.6% on 47% auto-accepted
Phase 2 · Size class	Random forest on outline geometry	330	80.3%	–	–	–
Phase 2 · Maturity stage	Ensemble: DINOv3 ViT-B/16 336 probe + EfficientNetV2-B0 260	463	70.0%	0.696	0.727	81.6% on 67% auto-accepted

Model choice, calibration temperature and review thresholds were all fixed on validation splits; each test set was used once for the numbers shown. Loss is categorical cross-entropy of the uncalibrated model with test-time augmentation. Colour uniformity, defects, skin and weight are measurement rules without labelled data (section 6).

Phase 1 · Variety classification

Random stratified 70/15/15 split of 17 varieties (Zenodo web photos, Alhamdan studio photos, and 200 grading-set photos per grading variety, kept in their grade split). Images of the original 10 varieties keep their earlier split.

99.5%
test accuracy

0.994
macro-F1

0.995
weighted-F1

0.102
test loss (cross-entropy)

99.8%
top-2 accuracy

411
test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	96.4%	0.975	0.248	not run
EfficientNetV2-B0 260	97.6%	0.982	0.172	96.1%
Ensemble: EfficientNetV2-B0 260 + MobileNetV3-Large 224	97.3%	0.981	0.202	not run
DINOv3 ViT-B/16 336 probe	99.0%	0.994	0.020	99.3%
DINOv3 ViT-L/16 336 probe	99.3%	0.987	0.019	not run
DINOv2 ViT-L/14 336 probe	99.0%	0.981	0.027	not run
Ensemble: DINOv3 ViT-B/16 336 probe + EfficientNetV2-B0 260	99.8%	0.998	0.083	99.5%
Ensemble: DINOv3 ViT-L/16 336 probe + DINOv2 ViT-L/14 336 probe + EfficientNetV2-B0 260	99.8%	0.998	0.061	not run
DINOv3 ViT-B/16 336 probe (background-augmented)	99.0%	0.994	0.020	99.3%
Ensemble: DINOv3 ViT-B/16 336 probe (background-augmented) + EfficientNetV2-B0 260 ✓ selected	99.5%	0.997	0.082	99.5%

Selected on validation: **Ensemble: DINOv3 ViT-B/16 336 probe (background-augmented) + EfficientNetV2-B0 260** (highest validation accuracy among candidates that pass the explainability sanity checks; ties go to the smaller model). CNNs: ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Probes (PyTorch, Apple GPU): frozen self-supervised DINOv3/DINOv2 backbone, [CLS, mean patch] embedding, logistic regression with C chosen on validation loss. Ensembles average probabilities. Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

Background shortcut found and fixed. The first DINOv3 probe scored 99.0 % on validation but only 82.1 % once the background was removed (the CNN: 96.8 → 95.6 % on test): the frozen backbone also encodes each variety's photo session (lighting, framing), and the probe used it. Its Grad-CAM showed the same, with heat on the background around the fruit. Training the probe on every image twice, as photographed and with the background removed (--bg-aug), keeps 99.0 % on original photos and reaches 98.8 % without background. It was chosen over the slightly higher but shortcut-prone ensemble (99.8 % vs 99.5 % validation) on this check, before the test set was scored.

Per-class results (test)

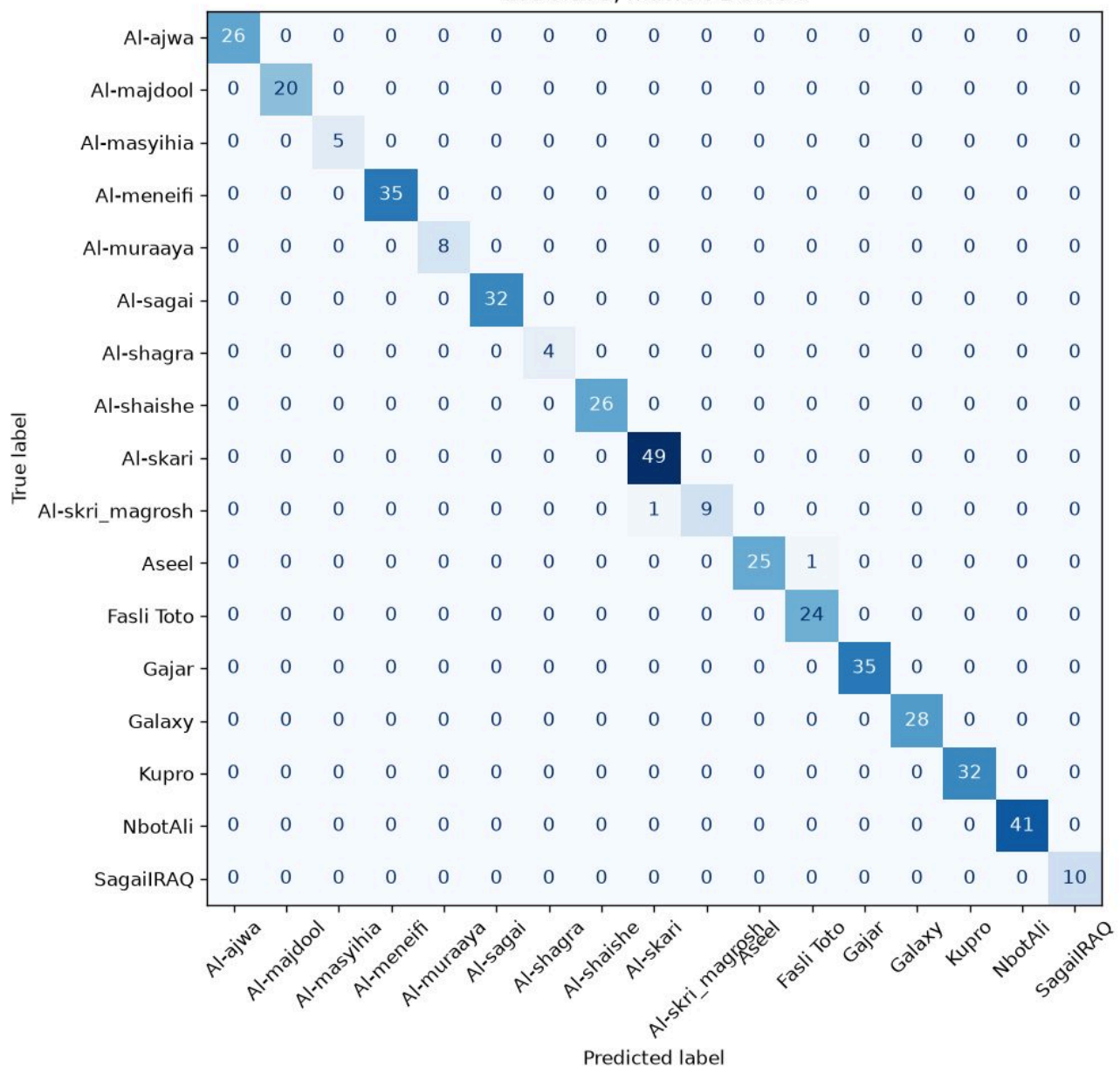
Class	Precision	Recall	F1	n
Al-ajwa	1.000	1.000	1.000	26
Al-majdool	1.000	1.000	1.000	20
Al-masyihia	1.000	1.000	1.000	5
Al-meneifi	1.000	1.000	1.000	35
Al-muraaya	1.000	1.000	1.000	8
Al-sagai	1.000	1.000	1.000	32
Al-shagra	1.000	1.000	1.000	4
Al-shaishe	1.000	1.000	1.000	26
Al-skari	0.980	1.000	0.990	49
Al-skri_magrosh	1.000	0.900	0.947	10
Aseel	1.000	0.962	0.980	26
Fasli Toto	0.960	1.000	0.980	24
Gajar	1.000	1.000	1.000	35
Galaxy	1.000	1.000	1.000	28
Kupro	1.000	1.000	1.000	32
NbotAli	1.000	1.000	1.000	41
SagailRAQ	1.000	1.000	1.000	10

Accuracy by source

Source / variety	Accuracy
alhamdan	100.0%
grading	99.1%
zenodo	98.5%

Confusion matrix (test)

test confusion matrix (probe_dinov3_b_bgaug+effv2b0_260_v2_tta)
acc 0.995, macro-F1 0.994



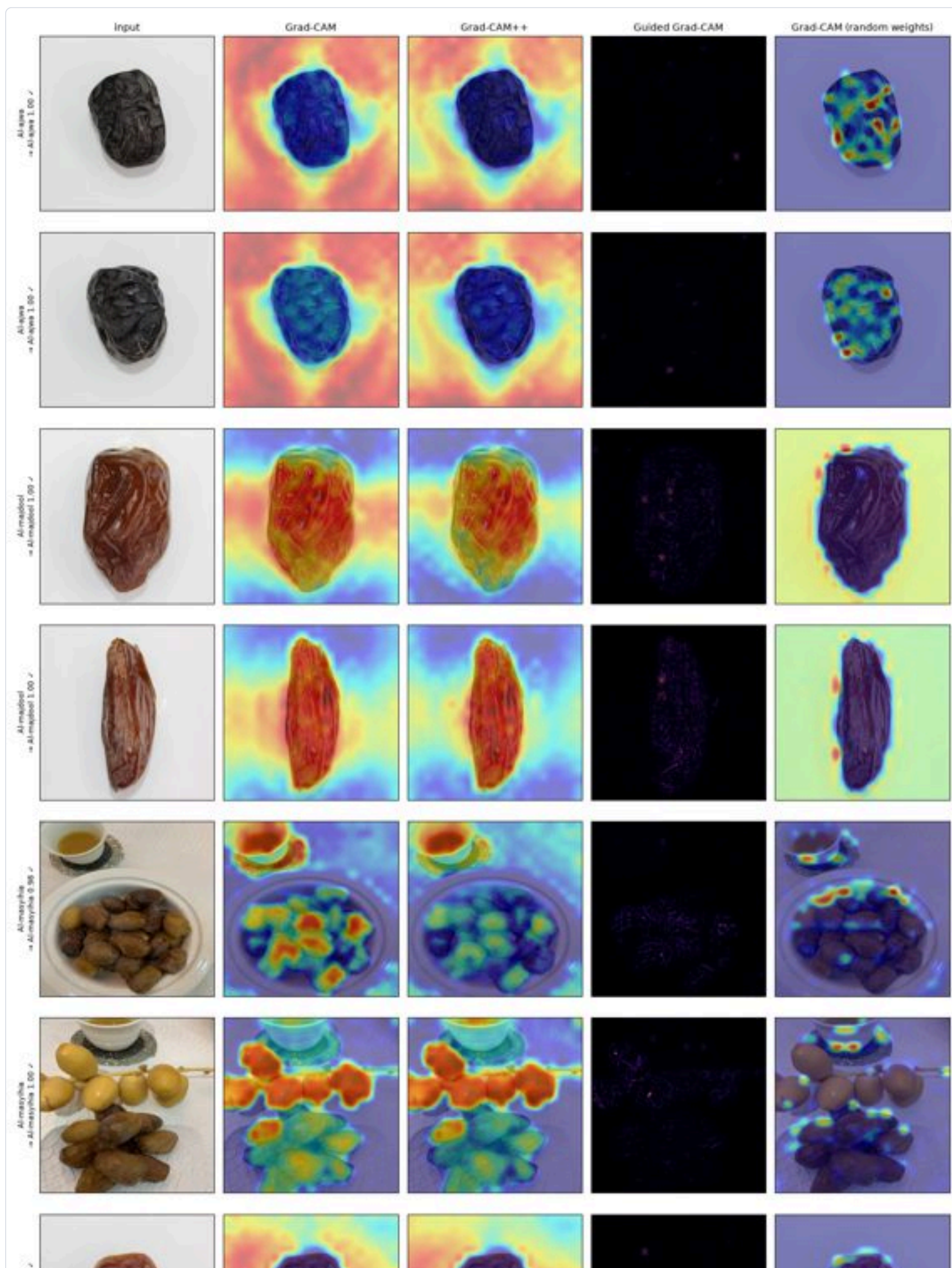
Calibration and manual-review threshold

Temperature scaling ($T = 1.0$) fitted on validation. Expected calibration error on test $0.076 \rightarrow \mathbf{0.076}$; test NLL $0.102 \rightarrow 0.102$. The review threshold (0.434) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **99.8%** of images at **99.5%** accuracy and sends 1 of 411 to manual review.

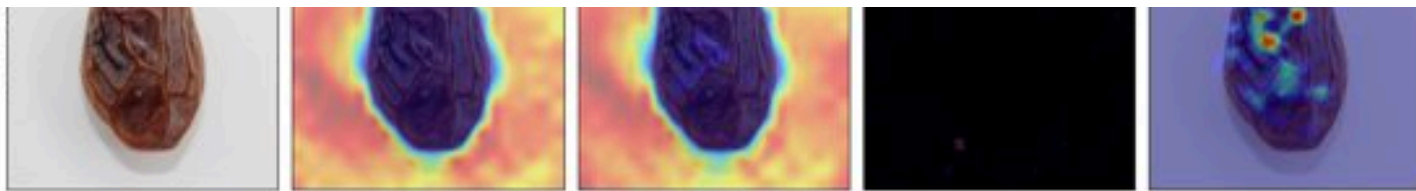
Confidence $\geq$	Auto-accepted	Accuracy on accepted
0.00	100.0%	99.5%
0.50	99.8%	99.5%
0.60	96.8%	99.5%
0.70	95.6%	99.8%
0.80	91.0%	100.0%
0.85	88.3%	100.0%
0.90	83.9%	100.0%
0.95	52.8%	100.0%
0.98	0.2%	100.0%

Explainability: Grad-CAM sanity checks (test split) · ensemble member DINOv3 ViT-B/16 336 probe (background-augmented)

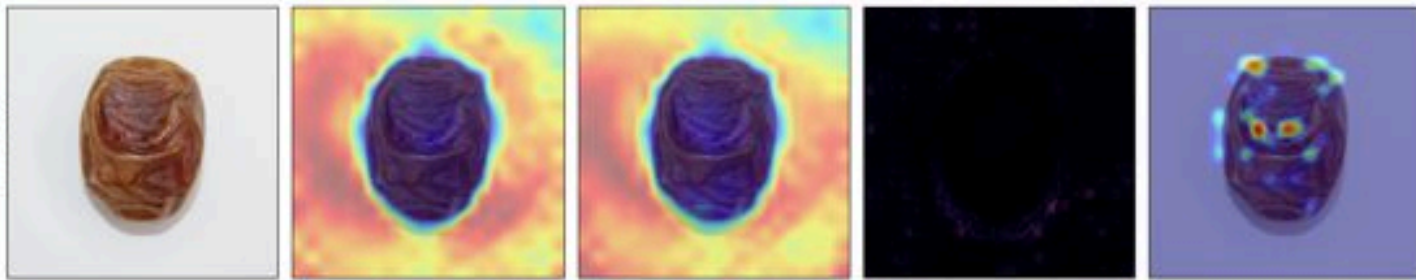
Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	-0.425	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.150 vs 0.057	pass
Share of heatmap energy on fruit pixels vs fruit share of image	39.0% vs 32.7%	pass
Test accuracy with the background removed vs as photographed (single view)	98.5% vs 99.3%	pass



Al-murashī
= Al-murashī 1.00 ✓



Al-murashī
= Al-murashī 1.00 ✓



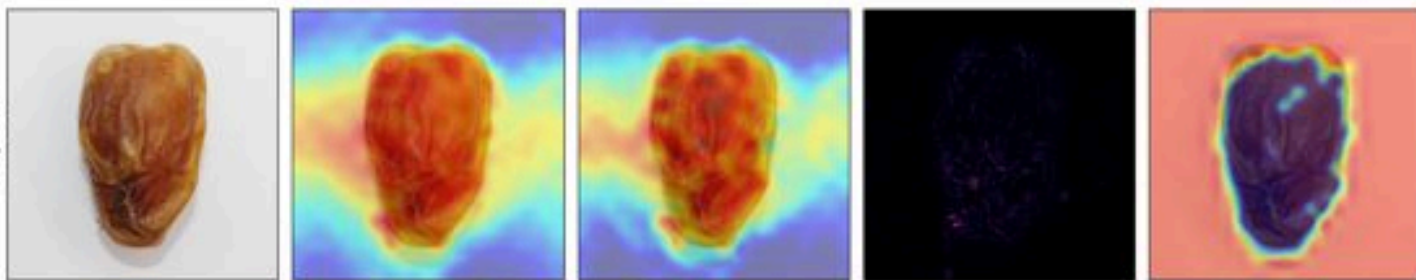
Al-murashī
= Al-murashī 0.98 ✓



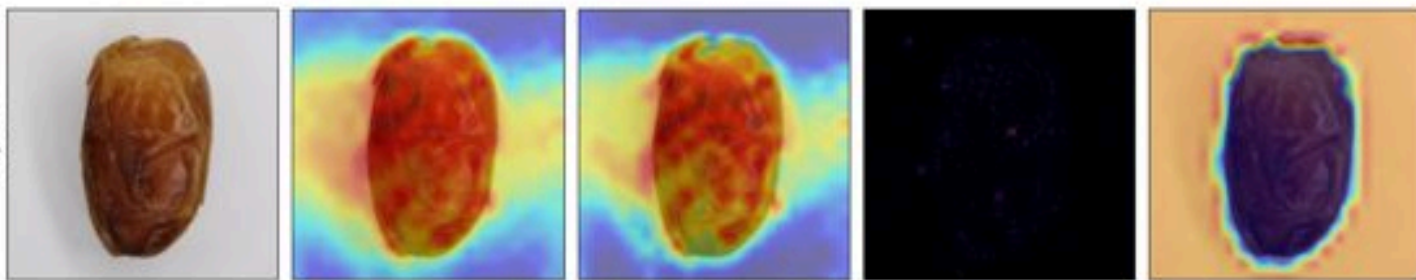
Al-murashī
= Al-murashī 1.00 ✓



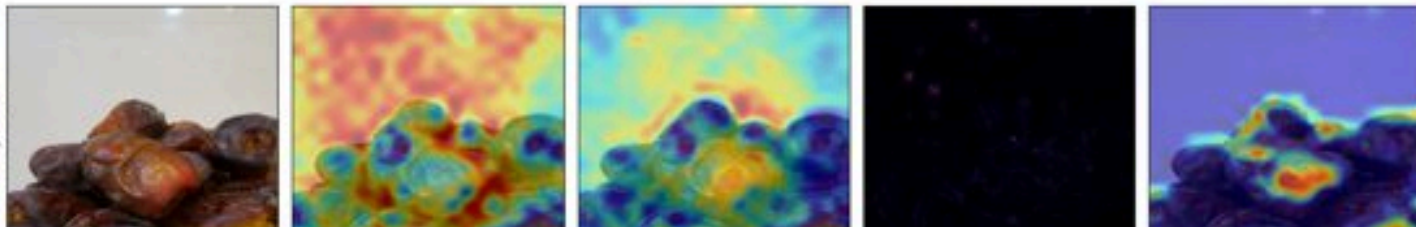
Al-hagāḡ
= Al-hagāḡ 1.00 ✓

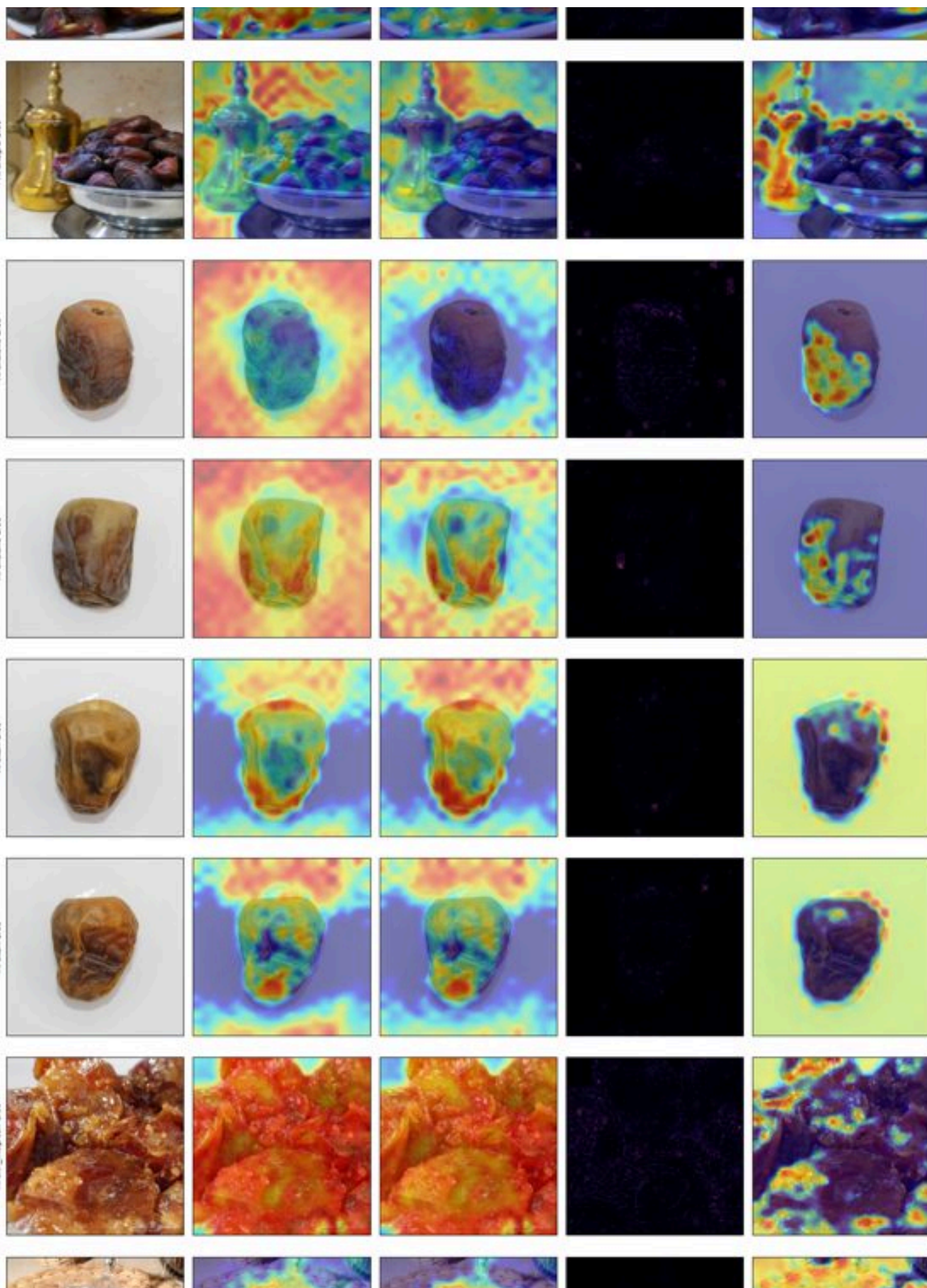


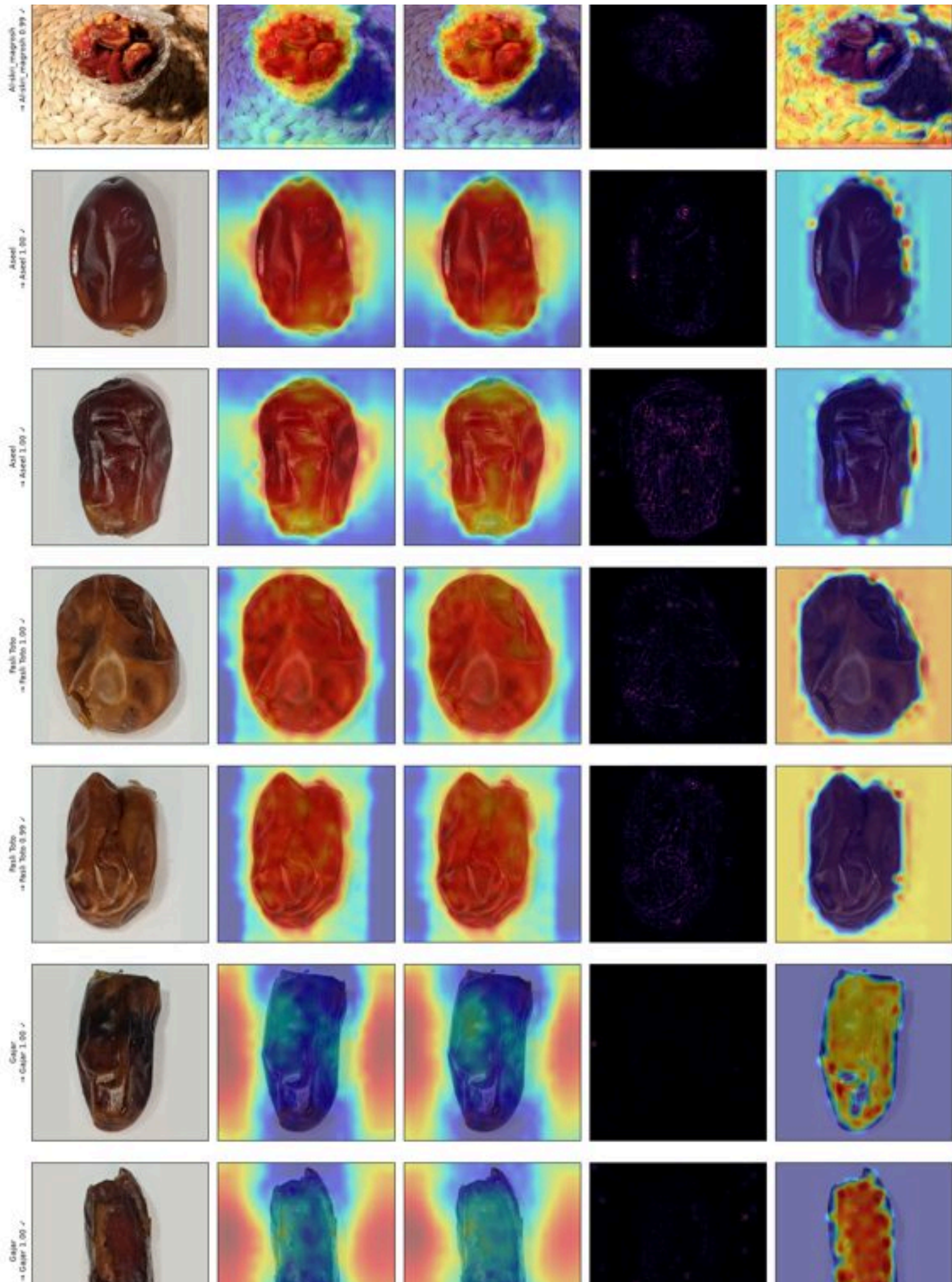
Al-hagāḡ
= Al-hagāḡ 1.00 ✓



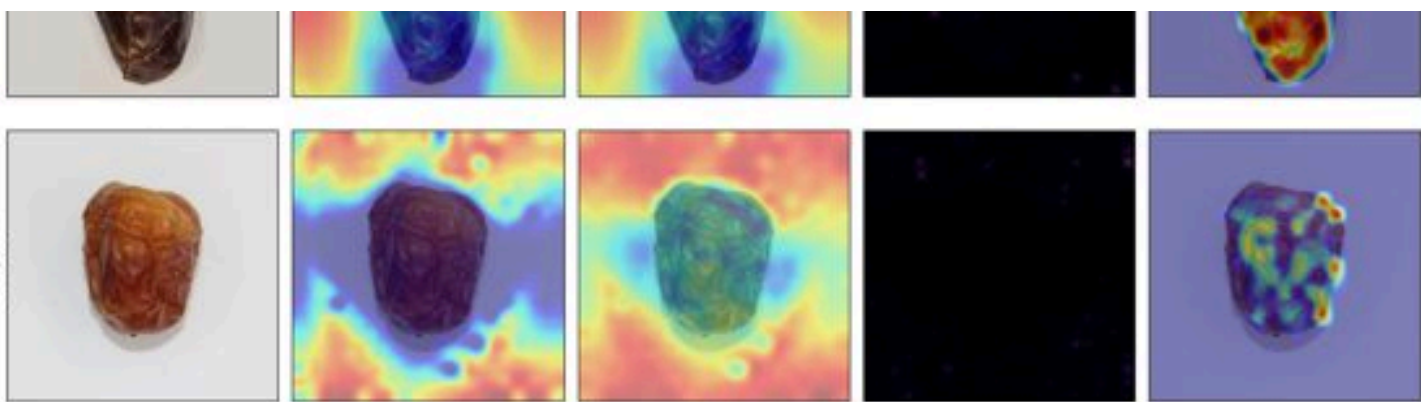
Al-hagāḡ
= Al-hagāḡ 1.00 ✓



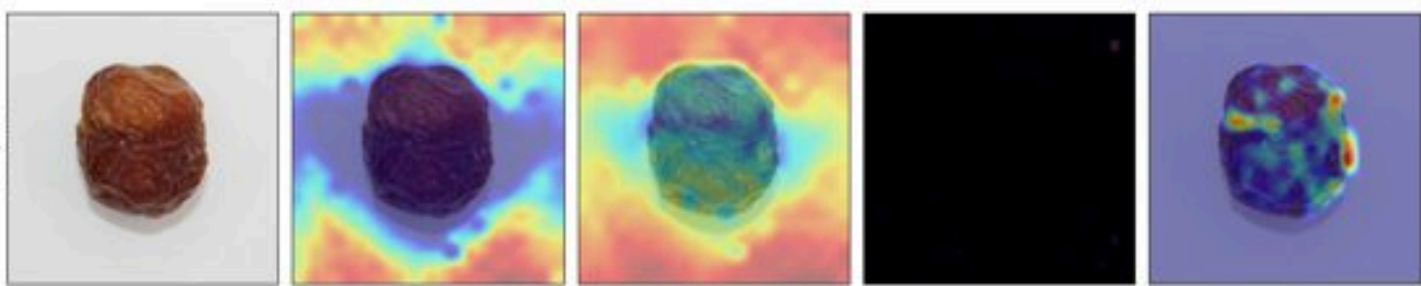




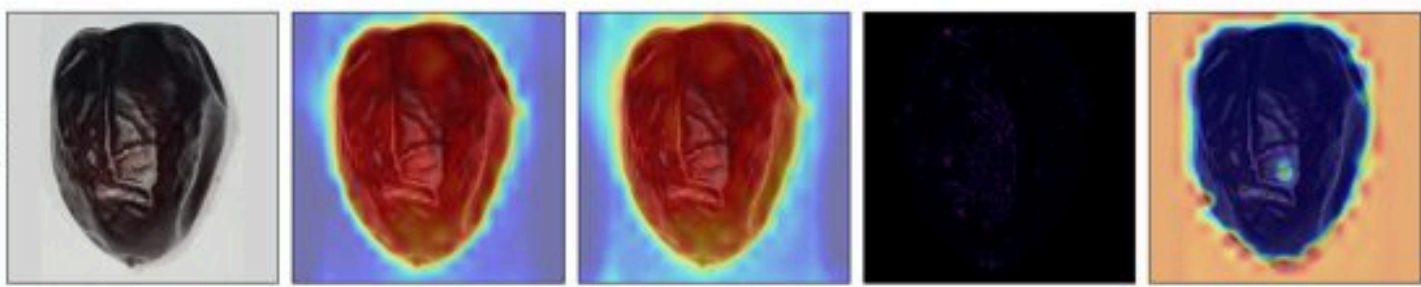
Galaxy
→ Galaxy 1.00 ✓



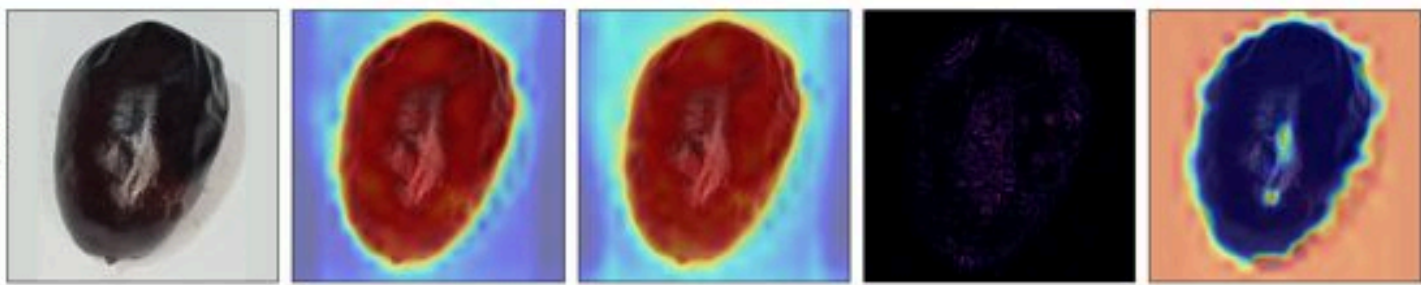
Galaxy
→ Galaxy 1.00 ✓



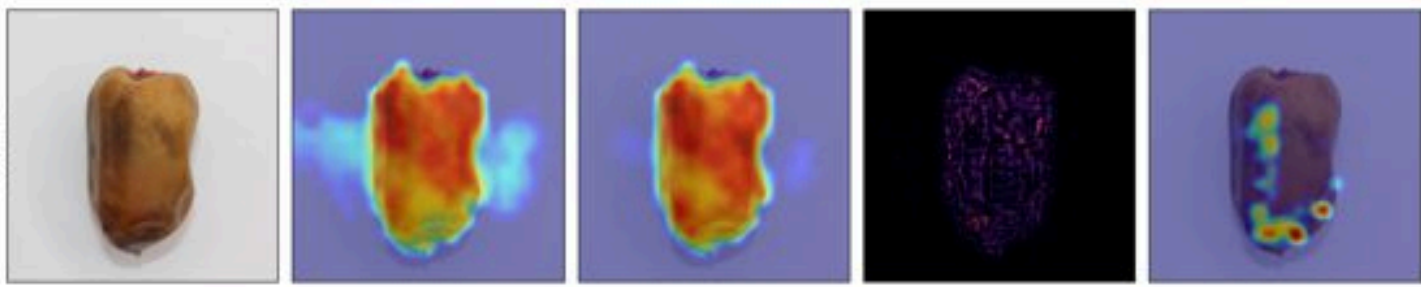
Kuiper
→ Kuiper 1.00 ✓



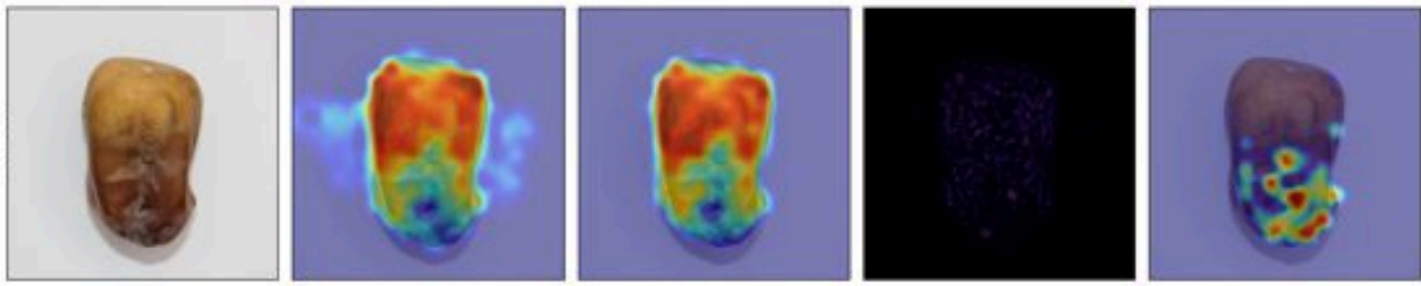
Kuiper
→ Kuiper 1.00 ✓

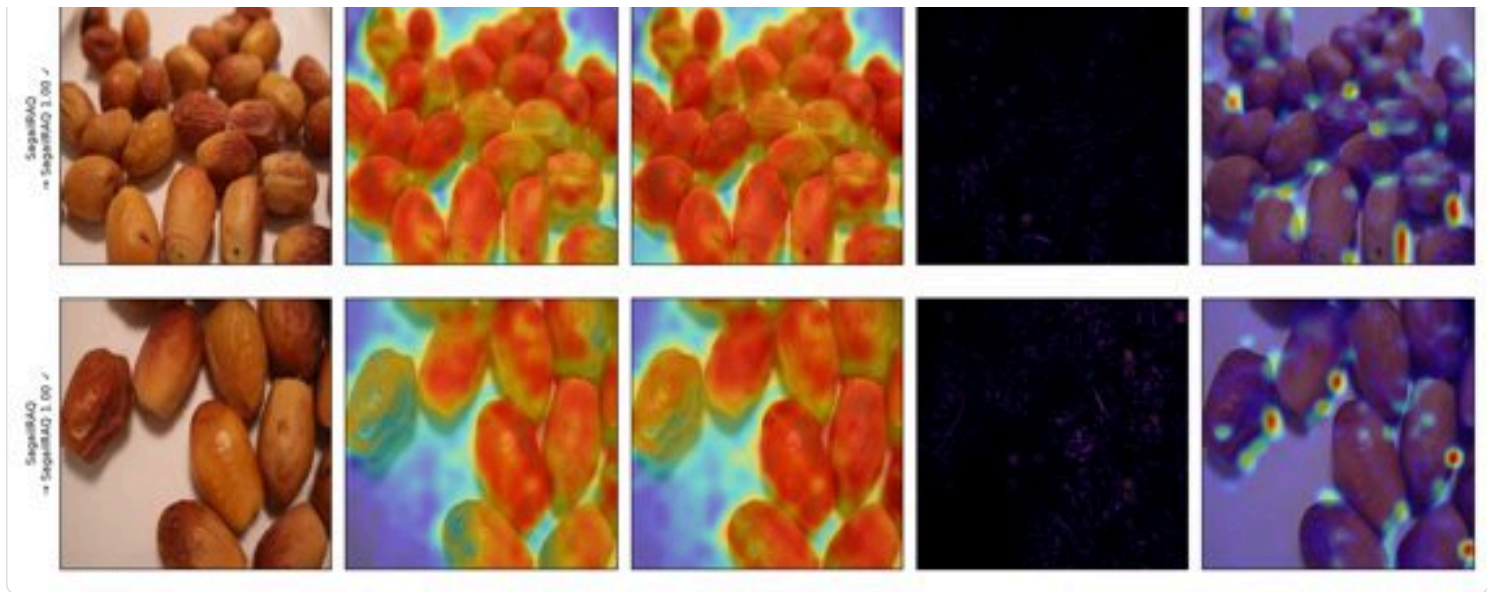


Neolith
→ Neolith 1.00 ✓



Neolith
→ Neolith 1.00 ✓

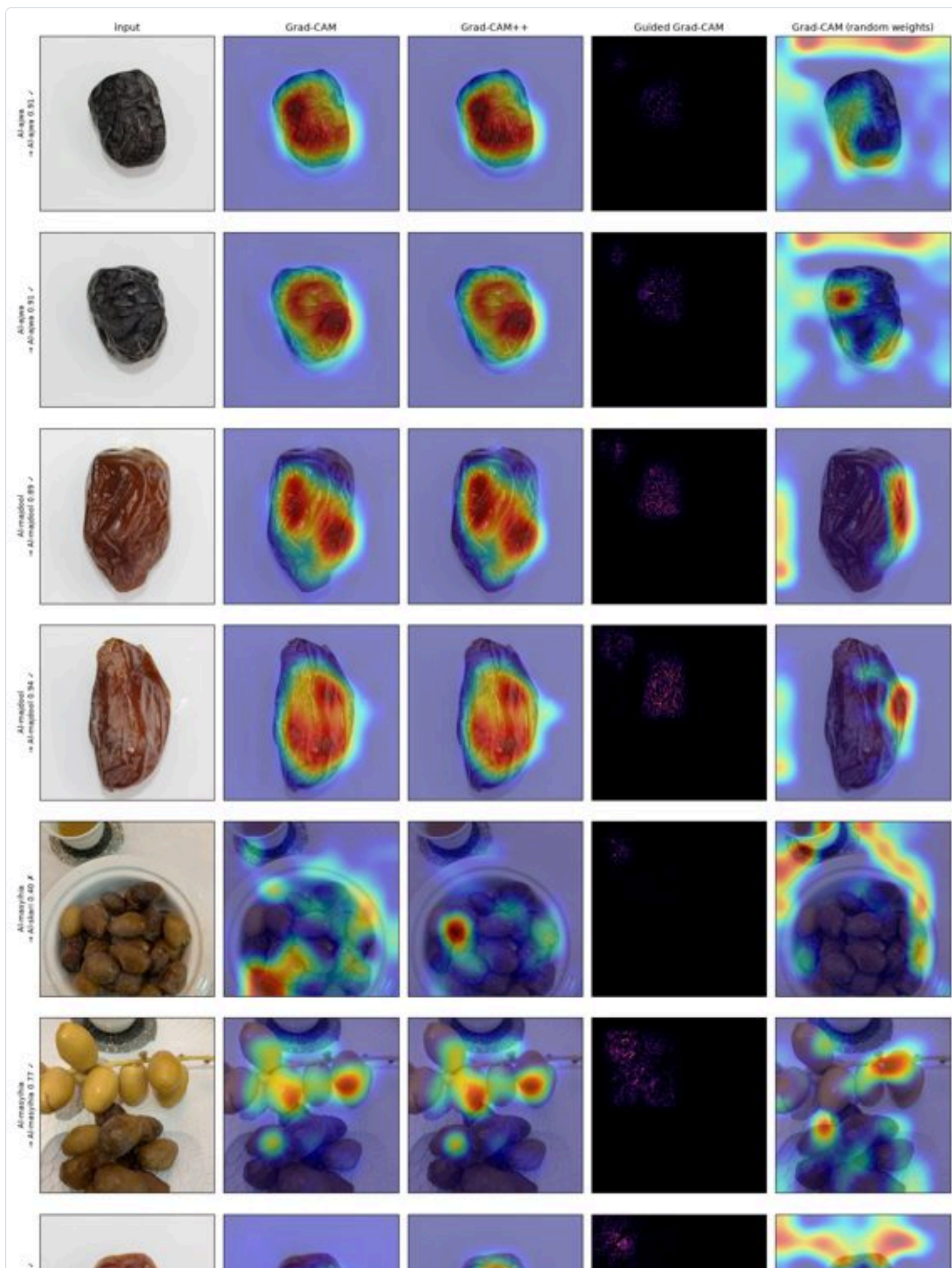




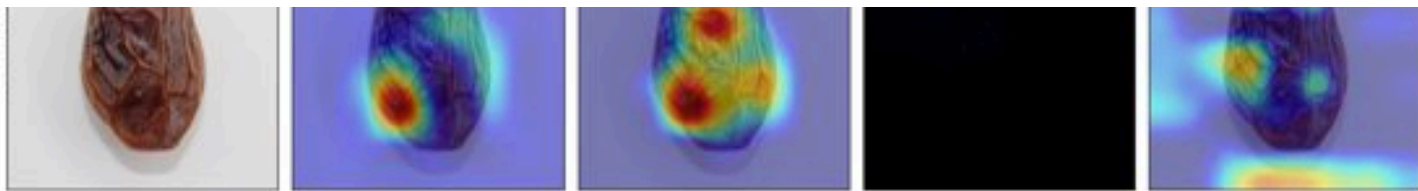
Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model. For the ViT, Grad-CAM runs on the final patch tokens (explain_vit.py).

Explainability: Grad-CAM sanity checks (test split) · ensemble member EfficientNetV2-B0 260

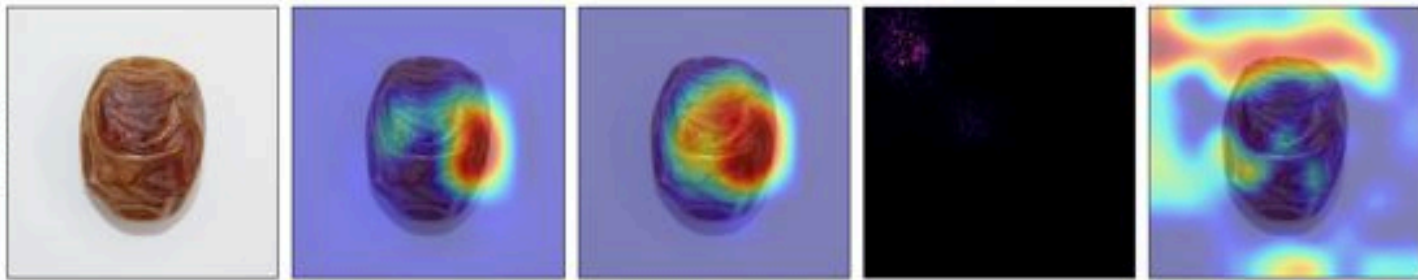
Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	-0.000	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.460 vs 0.289	pass
Share of heatmap energy on fruit pixels vs fruit share of image	75.9% vs 31.6%	pass



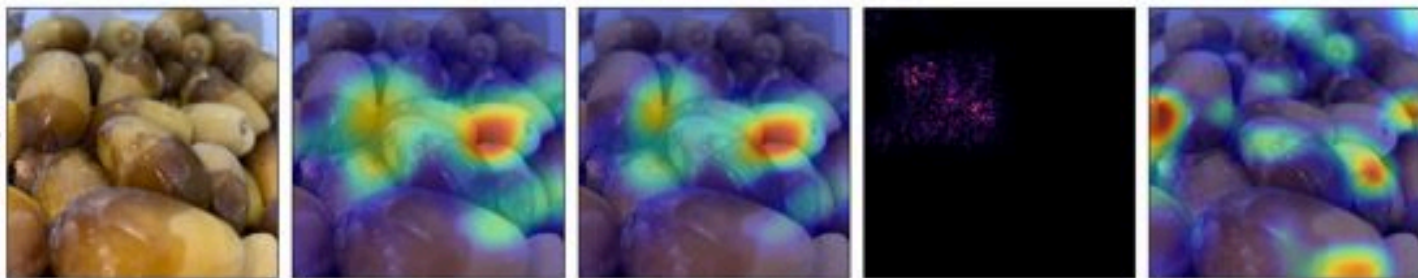
Al-muradhi
= Al-muradhi 0.85



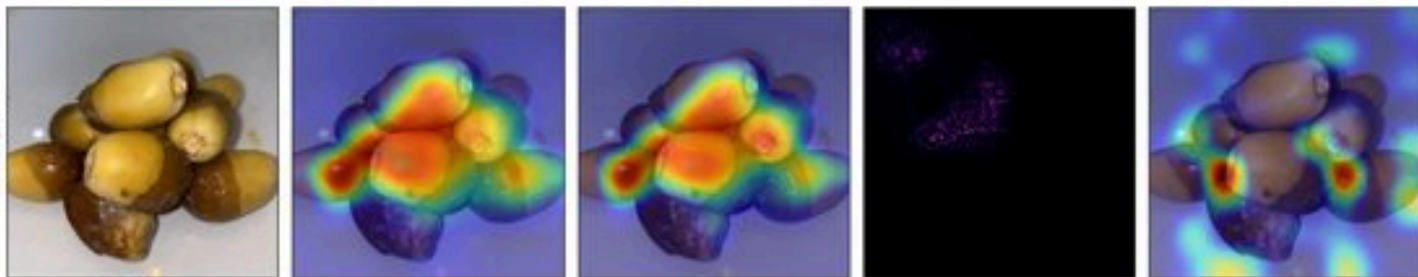
Al-muradhi
= Al-muradhi 0.64



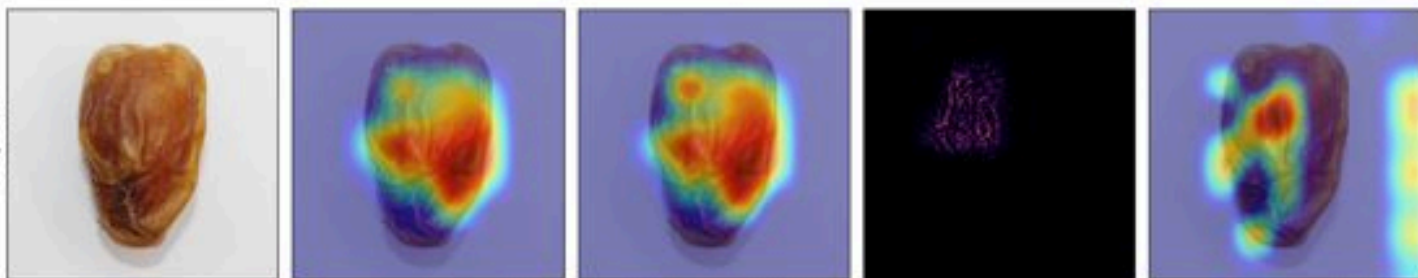
Al-muradhi
= Al-muradhi 0.71



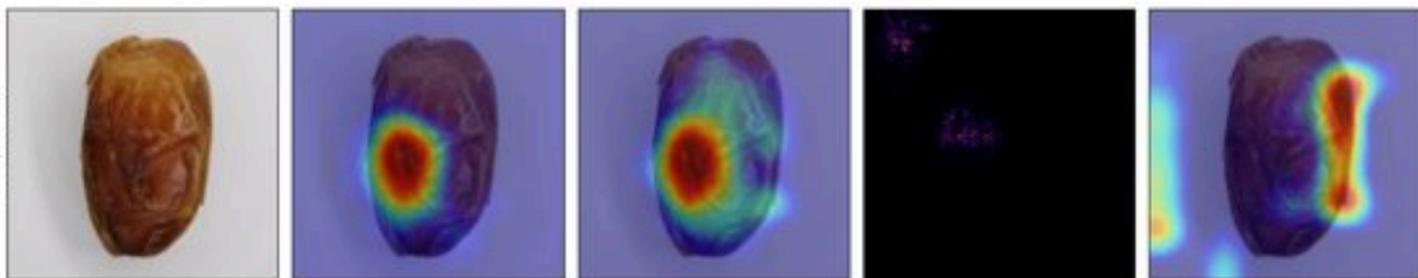
Al-muradhi
= Al-muradhi 0.53



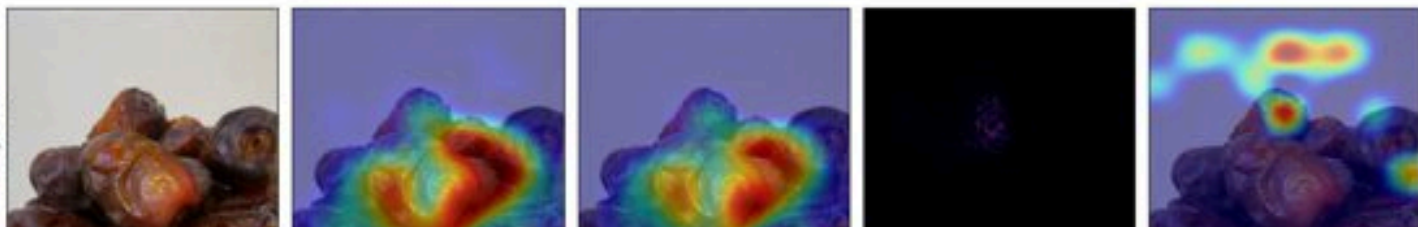
Al-saghi
= Al-saghi 0.91

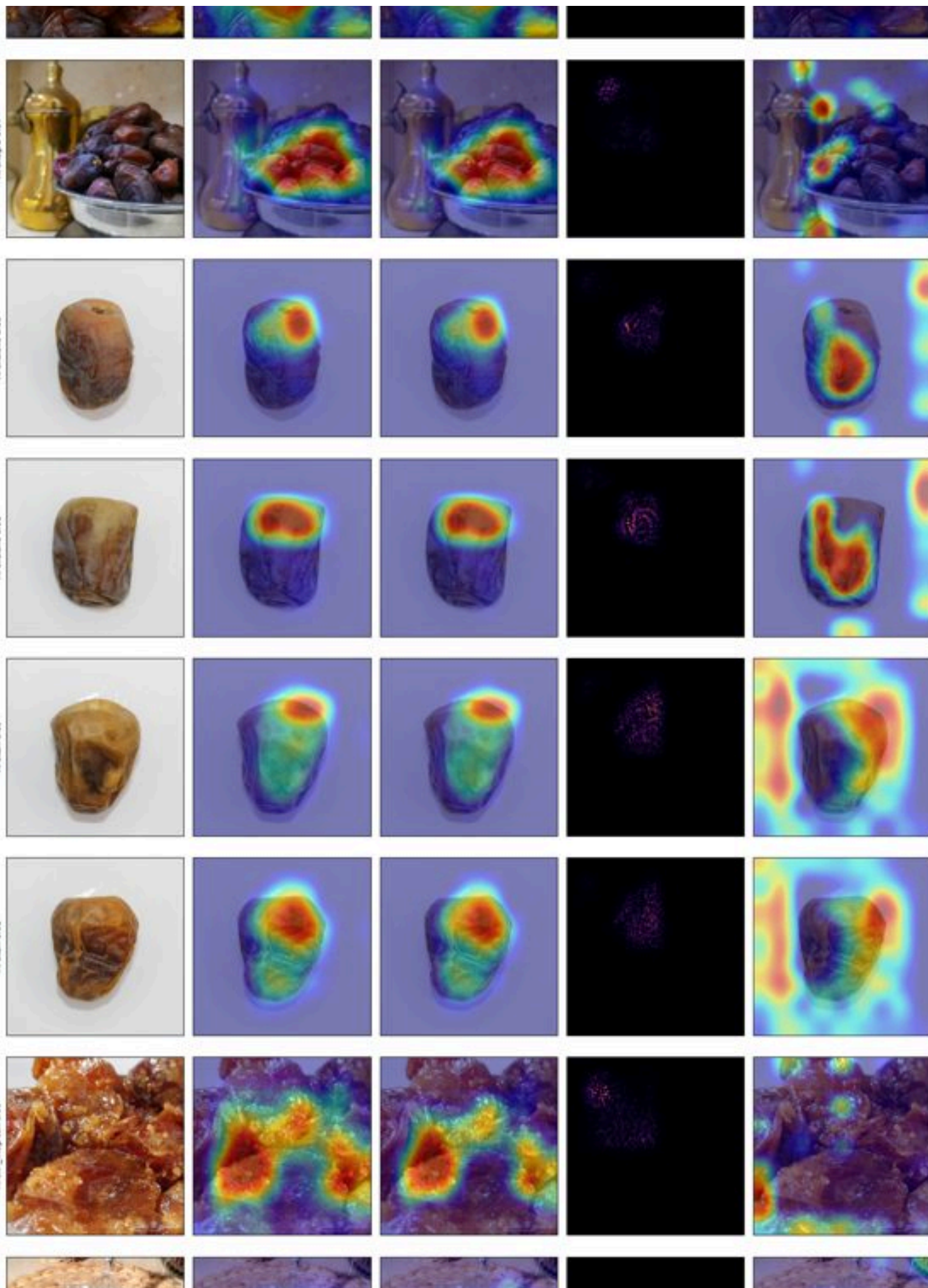


Al-saghi
= Al-saghi 0.60

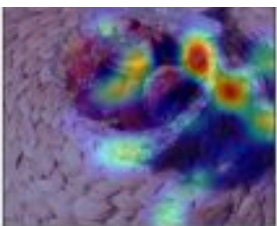
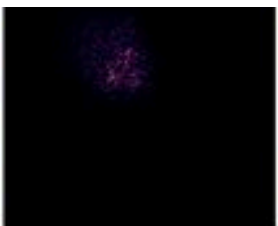
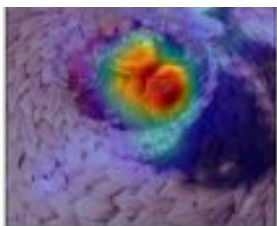
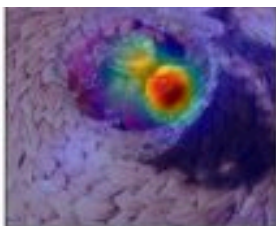


Al-saghi
= Al-saghi 0.91

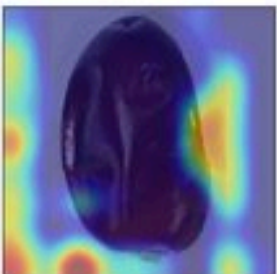
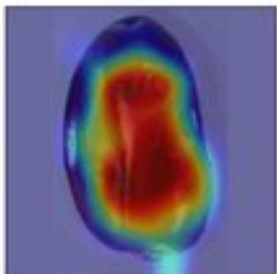
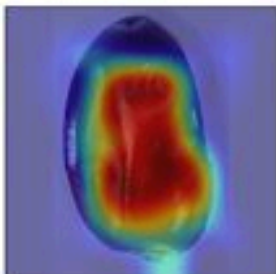
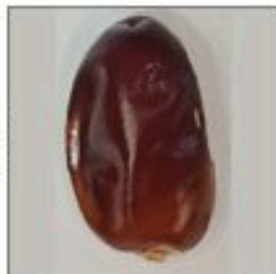




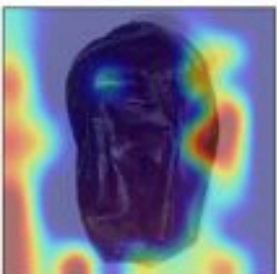
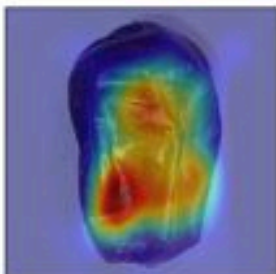
Al-Iski magnon
→ Al-Iski 0.23 ✓



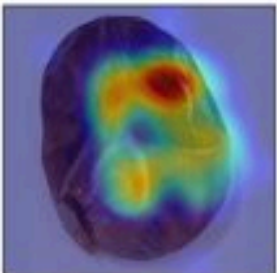
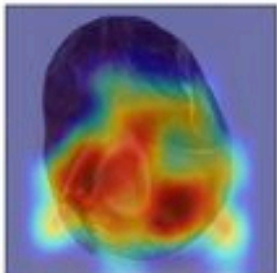
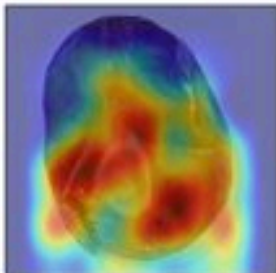
Al-Iski
→ Al-Iski 0.92 ✓



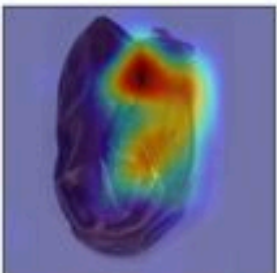
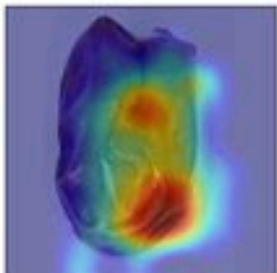
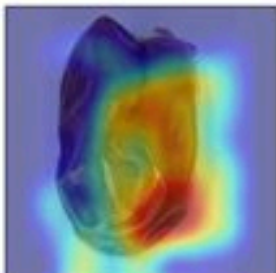
Al-Iski
→ Al-Iski 0.88 ✓



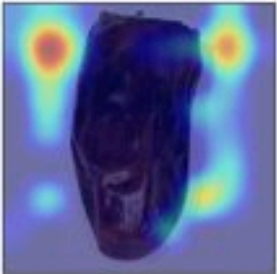
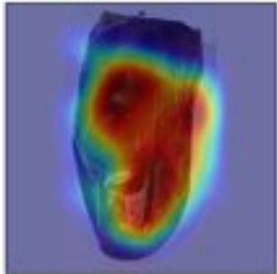
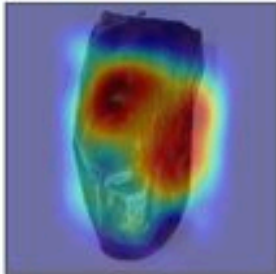
Flash Tolo
→ Flash Tolo 0.93 ✓



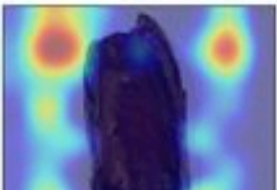
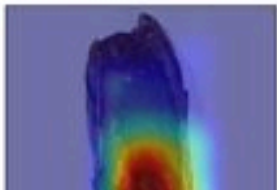
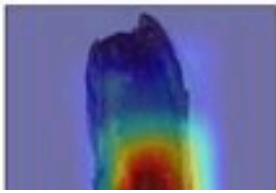
Flash Tolo
→ Flash Tolo 0.90 ✓



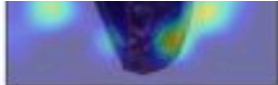
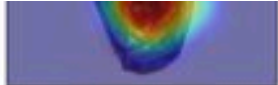
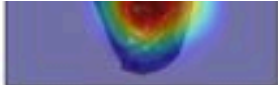
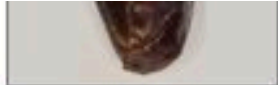
Galax
→ Galax 0.88 ✓



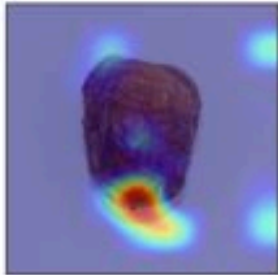
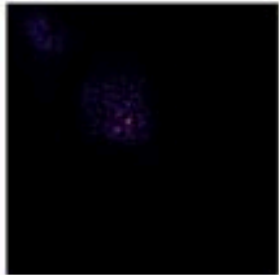
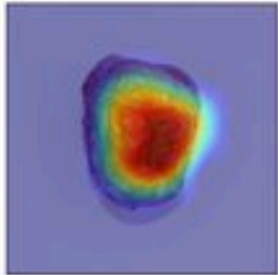
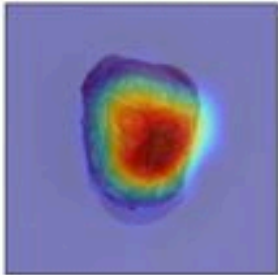
Galax
→ Galax 0.96 ✓



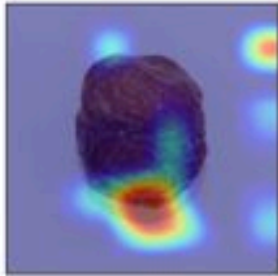
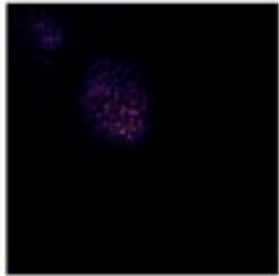
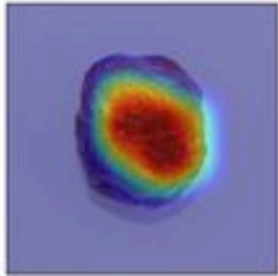
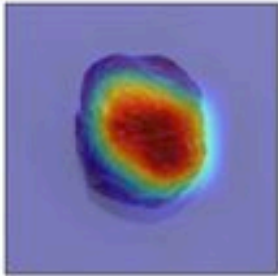
Galaxy
→ Galaxy 0.90 ✓



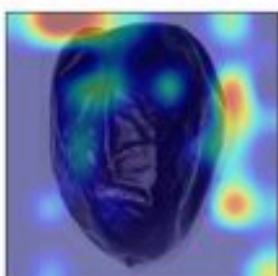
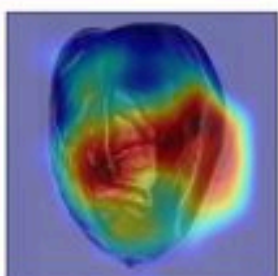
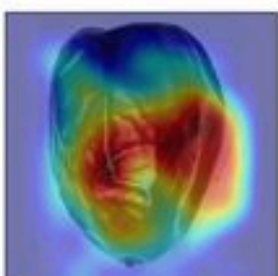
Galaxy
→ Galaxy 0.88 ✓



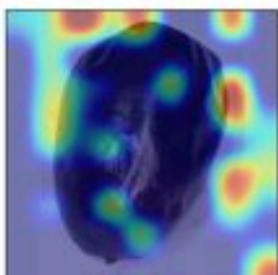
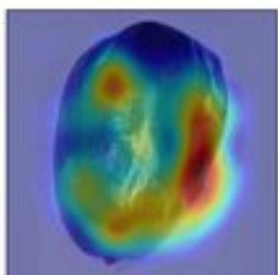
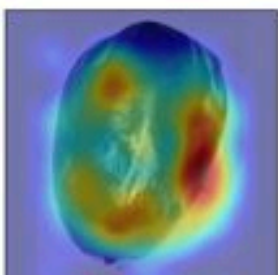
Kupar
→ Kupar 0.92 ✓



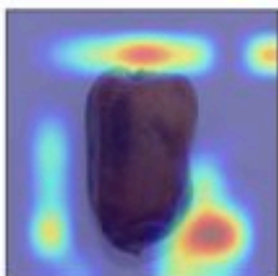
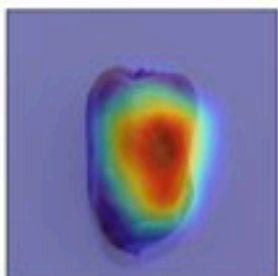
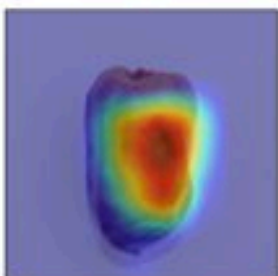
Kupar
→ Kupar 0.93 ✓

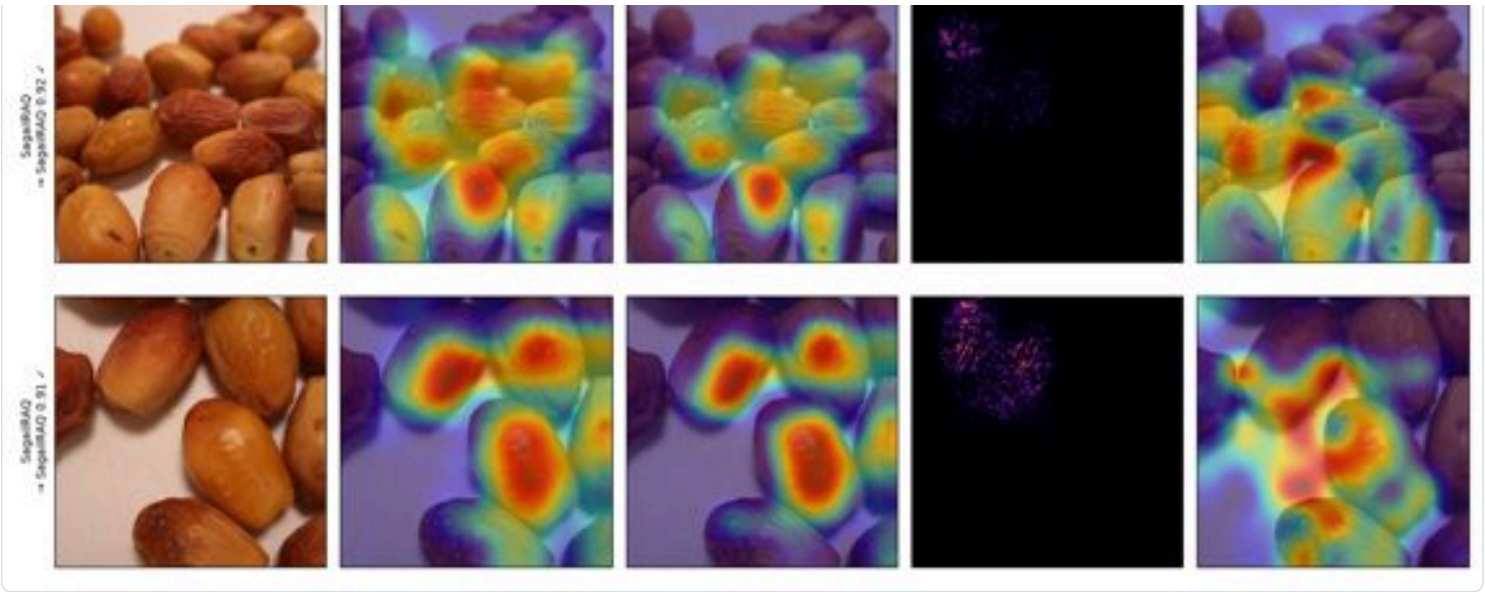


Morokk
→ Morokk 0.94 ✓



Morokk
→ Morokk 0.93 ✓





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
EfficientNetV2-B0 260	11.83	71.38	100.0% (n=200)	0.0076	26.0 ms

Phase 1 · Colour, shape and texture per variety

Extracted from the segmented fruit (GrabCut / plain-background mask) of every training image; the table shows each variety's median. `python src/predict.py <image>` reports the same features for one photo next to the predicted variety's typical values, with the Grad-CAM heatmap. Shape values are scale-free ratios; hue is the circular mean of skin pixels; wrinkle index is the edge density inside the fruit.

Variety	Colour	Hue °	Brightness	Saturation	Colour uniformity	Shape	Aspect ratio	Circularity	Solidity	GLCM homogeneity	GLCM contrast	LBP entropy	Wrinkle index
Al-ajwa	black / very dark brown	14	0.26	0.07	0.40	oval	1.50	0.81	0.985	0.551	5.07	3.19	0.103
Al-majdool	amber brown	15	0.52	0.51	0.37	oblong / elongated	1.75	0.76	0.987	0.552	5.24	3.17	0.112
Al-masyihia	amber / golden yellow	32	0.57	0.55	0.27	round	1.25	0.49	0.879	0.656	3.50	3.03	0.067
Al-meneifi	amber brown	17	0.54	0.58	0.34	oblong / elongated	1.65	0.79	0.989	0.662	2.09	3.15	0.038
Al-muraaya	golden brown	35	0.54	0.65	0.21	oval	1.40	0.56	0.893	0.719	2.18	3.08	0.031
Al-sagai	amber brown	22	0.57	0.64	0.33	oblong / elongated	1.77	0.75	0.986	0.649	2.27	3.16	0.040
Al-shagra	amber brown	14	0.45	0.55	0.24	oval	1.39	0.56	0.904	0.612	4.16	2.97	0.086
Al-shaishe	amber brown	24	0.51	0.46	0.35	oblong / elongated	1.60	0.80	0.988	0.658	2.08	3.17	0.036
Al-skari	amber brown	25	0.56	0.59	0.31	oval	1.44	0.80	0.986	0.657	1.96	3.19	0.029
Al-skri_magrosh	amber brown	25	0.62	0.64	0.23	oval	1.28	0.61	0.932	0.583	5.46	3.04	0.106
Aseel	dark red-brown	12	0.40	0.57	0.40	oblong / elongated	1.68	0.76	0.985	0.688	2.83	3.15	0.056
Fasli Toto	amber brown	20	0.43	0.64	0.39	oval	1.56	0.73	0.978	0.735	1.92	3.10	0.034
Gajar	dark brown	14	0.27	0.42	0.41	very elongated	2.23	0.69	0.980	0.640	4.23	3.17	0.077
Galaxy	amber brown	15	0.50	0.56	0.33	oval	1.39	0.81	0.985	0.626	2.79	3.18	0.053
Kupro	black / very dark brown	331	0.23	0.21	0.38	oval	1.43	0.78	0.982	0.703	2.78	3.17	0.053
NbotAli	amber brown	23	0.56	0.62	0.34	oblong / elongated	1.63	0.76	0.984	0.704	1.46	3.15	0.014
SagailIRAQ	amber brown	22	0.65	0.77	0.31	oblong / elongated	1.66	0.48	0.853	0.738	1.37	2.84	0.022

Phase 2 · Quality grade (1–3)

Stratified split of the Kaggle grading set (4 varieties). Aseel and Fasli Toto only have Grade-1, so the informative scores are Gajar and Kupro.

84.5%
test accuracy

0.813
macro-F1

0.848
weighted-F1

0.371
test loss (cross-entropy)

98.9%
top-2 accuracy

451
test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	87.4%	0.853	0.349	84.9%
EfficientNetV2-B0 260	88.2%	0.860	0.332	84.7%
Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260 ✓ selected	89.8%	0.881	0.330	84.5%
DINOv3 ViT-B/16 336 probe	82.5%	0.800	0.386	not run
DINOv3 ViT-L/16 336 probe	80.5%	0.775	0.414	not run
DINOv3 ViT-B/16 336 fine-tuned	87.4%	0.842	0.387	not run
Ensemble: DINOv3 ViT-B/16 336 probe + MobileNetV3-Large 224 + EfficientNetV2-B0 260	88.2%	0.859	0.331	not run
Ensemble: DINOv3 ViT-B/16 336 fine-tuned + MobileNetV3-Large 224 + EfficientNetV2-B0 260	87.6%	0.848	0.349	not run

Selected on validation: **Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260** (highest validation accuracy among candidates that pass the explainability sanity checks; ties go to the smaller model). CNNs: ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Probes (PyTorch, Apple GPU): frozen self-supervised DINOv3/DINOv2 backbone, [CLS, mean patch] embedding, logistic regression with C chosen on validation loss. Ensembles average probabilities. Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

Modern backbones did not beat the CNN ensemble here: a frozen DINOv3 probe (80–83 %) misses the small surface defects that separate the grades, and fully fine-tuning DINOv3-B at 336 px (layer-wise LR decay, fp16 on the Apple GPU) reached 87.4 % on validation vs 89.8 %; ensembling it with the CNNs did not help either. Three different model families plateauing at 87–90 % points to the subjective Grade-1/Grade-2 boundary in the labels as the limit.

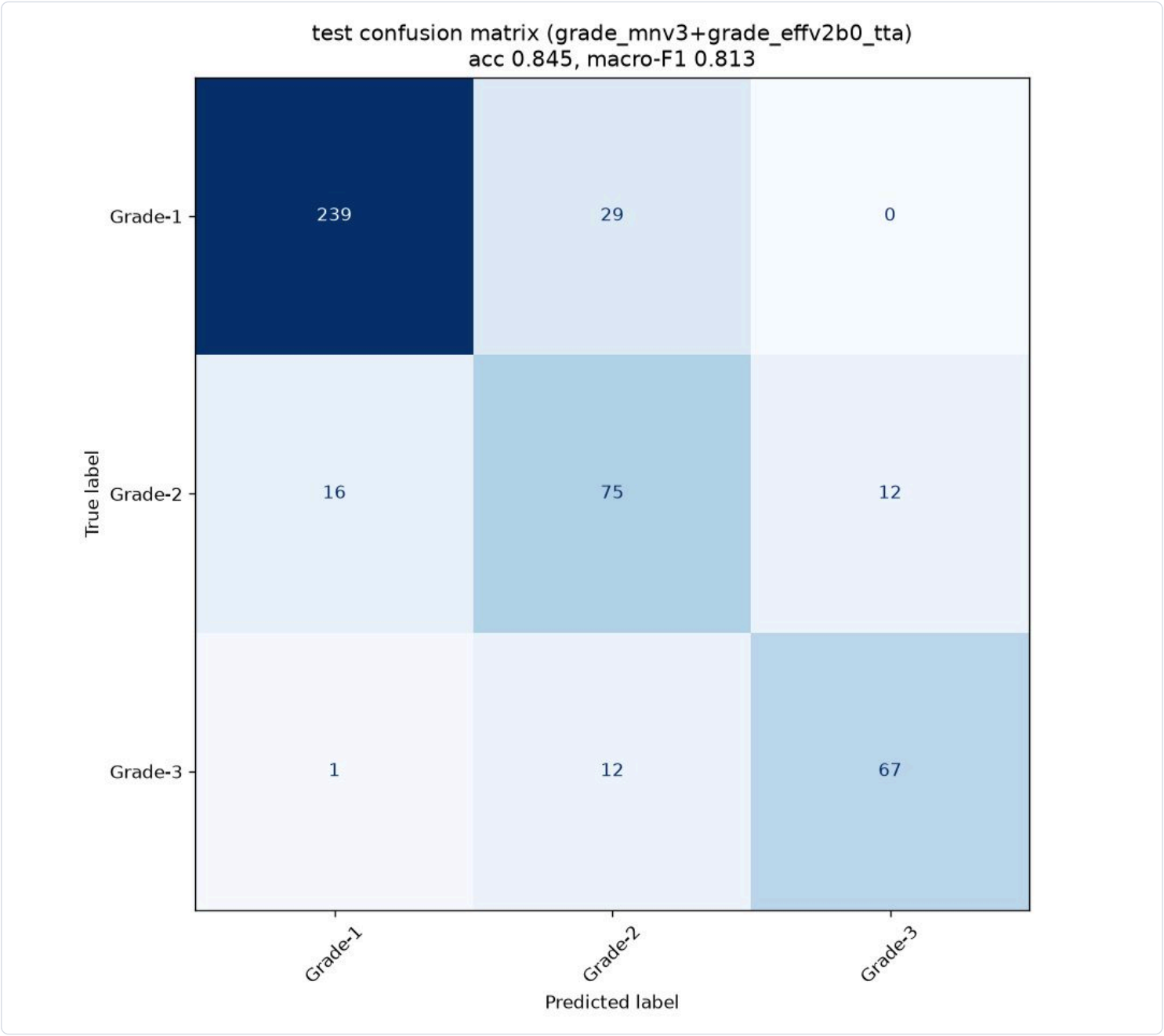
Per-class results (test)

Class	Precision	Recall	F1	n
Grade-1	0.934	0.892	0.912	268
Grade-2	0.647	0.728	0.685	103
Grade-3	0.848	0.838	0.843	80

Accuracy by source

Source / variety	Accuracy
Aseel	100.0%
Fasli-Toto	100.0%
Gajar	72.2%
Kupro	87.6%

Confusion matrix (test)



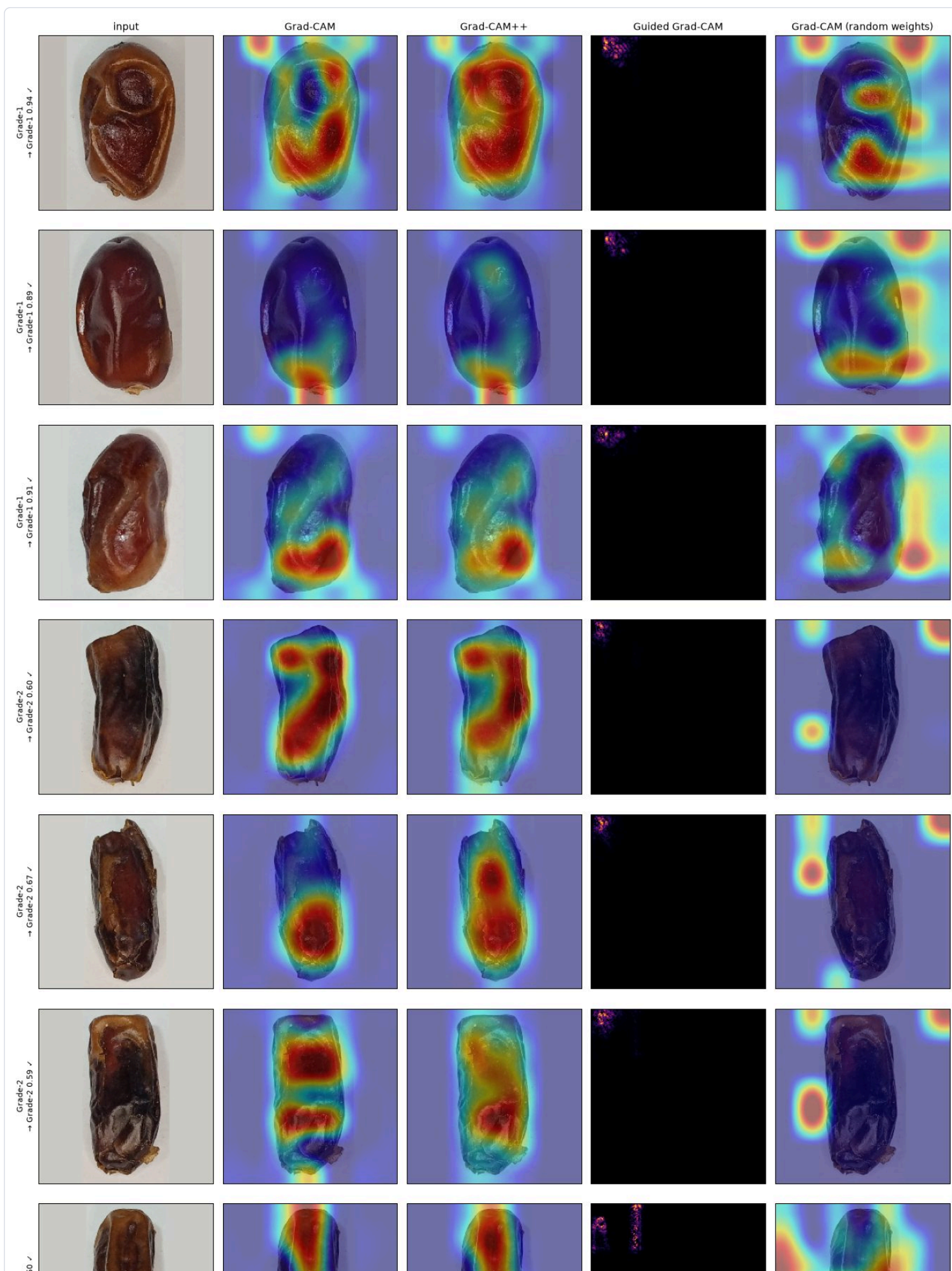
Calibration and manual-review threshold

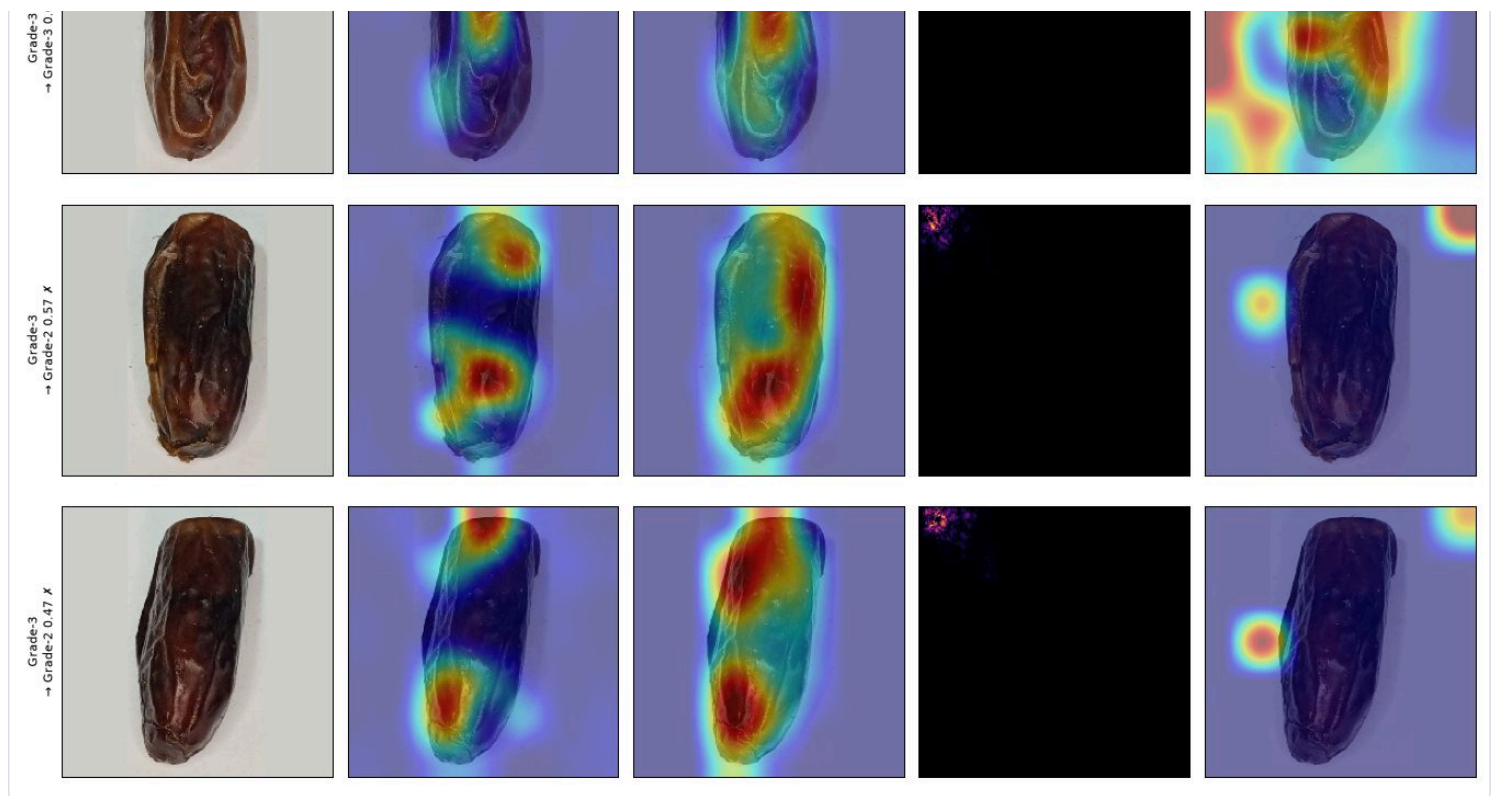
Temperature scaling ($T = 0.5486$) fitted on validation. Expected calibration error on test $0.080 \rightarrow 0.041$; test NLL $0.371 \rightarrow 0.343$. The review threshold (0.973) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **47.0%** of images at **97.6%** accuracy and sends 239 of 451 to manual review.

Confidence $\geq$	Auto-accepted	Accuracy on accepted
0.00	100.0%	84.5%
0.50	99.8%	84.7%
0.60	91.8%	88.6%
0.70	84.5%	91.9%
0.80	75.2%	95.6%
0.85	70.3%	95.9%
0.90	65.0%	97.3%
0.95	54.1%	98.0%
0.98	42.6%	98.4%

Explainability: Grad-CAM sanity checks (test split) · ensemble member MobileNetV3-Large 224

Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	-0.022	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.245 vs 0.021	pass
Share of heatmap energy on fruit pixels vs fruit share of image	82.4% vs 38.6%	pass





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
MobileNetV3-Large 224	6.0	36.36	100.0% (n=200)	0.013	13.8 ms
EfficientNetV2-B0 260	11.79	71.17	100.0% (n=200)	0.003	68.5 ms

Phase 2 · Size class (Small / Medium / Large)

80.3%

test accuracy (3 varieties with 3 sizes)

Per-variety random forest on the measured fruit outline: $\sqrt{\text{area}}$, length and width. Chosen on the validation split over logistic regression, gradient boosting and area-only cut points (val 0.839 vs 0.805–0.833). Size classes are relative to the variety (a Small Gajar is longer than a Large Aseel). Pixel models only hold for the grading-set camera rig; with an ArUco marker in the photo, size is measured in mm and classified with mm cut points.

Variety	Test accuracy	n
Aseel	98.6%	73
Fasli Toto	89.8%	59
Gajar	70.7%	198
Kupro	only Large in dataset	–

Aseel

	pred Small	pred Medium	pred Large
true Small	25	0	0
true Medium	1	24	0
true Large	0	0	23

Fasli Toto

	pred Small	pred Medium	pred Large
true Small	11	2	0
true Medium	2	19	2
true Large	0	0	23

Gajar

	pred Small	pred Medium	pred Large
true Small	67	12	4
true Medium	32	14	5
true Large	0	5	59

Gajar Small and Medium overlap almost completely in measured size (median $\sqrt{\text{area}}$ 347 vs 355 px), so the labels themselves cap accuracy for Gajar; more model capacity does not help (all four methods: 0.66–0.71).

Phase 2 · Maturity stage

Test = the whole Moroccan variety Kholt, never seen in training (each stage was shot in one session, so a random split would leak session lighting), plus held-out Alhamdan single fruits.

70.0%
test accuracy

0.696
macro-F1

0.694
weighted-F1

0.727
test loss (cross-entropy)

91.8%
top-2 accuracy

463
test images

Model selection

Candidate	Val accuracy	Val macro-F1	Val loss	Test accuracy (for disclosure)
MobileNetV3-Large 224	93.2%	0.931	0.279	70.0%
EfficientNetV2-B0 260	95.6%	0.956	0.206	67.8%
Ensemble: MobileNetV3-Large 224 + EfficientNetV2-B0 260	95.6%	0.956	0.229	not run
DINOv3 ViT-B/16 336 probe	95.3%	0.953	0.160	70.0%
DINOv3 ViT-L/16 336 probe	95.6%	0.957	0.112	not run
DINOv2 ViT-L/14 336 probe	95.6%	0.957	0.128	not run
Ensemble: DINOv3 ViT-B/16 336 probe + EfficientNetV2-B0 260 ✓ selected	97.3%	0.973	0.161	70.0%
Ensemble: DINOv3 ViT-L/16 336 probe + DINOv2 ViT-L/14 336 probe + EfficientNetV2-B0 260	96.9%	0.970	0.124	not run

Selected on validation: **Ensemble: DINOv3 ViT-B/16 336 probe + EfficientNetV2-B0 260** (highest validation accuracy among candidates that pass the explainability sanity checks; ties go to the smaller model). CNNs: ImageNet weights, frozen-backbone head training then whole-backbone fine-tuning (AdamW, cosine schedule, label smoothing 0.1). Probes (PyTorch, Apple GPU): frozen self-supervised DINOv3/DINOv2 backbone, [CLS, mean patch] embedding, logistic regression with C chosen on validation loss. Ensembles average probabilities. Test scores use 4-flip test-time augmentation. The test column is shown for transparency only and was not used to choose.

Validation (seen varieties, ~95–97 %) does not predict accuracy on a new variety (~65–70 %). A leave-one-variety-out check inside train+val gives the DINOv3-B probe 55–76 % on each held-out Moroccan variety, the same level as the CNN on the test variety: a stronger backbone does not solve new-variety ripeness, per-fruit stage labels would. On studio photos of single fruits (Alhamdan, Rutab vs Tamar) it is 100 %.

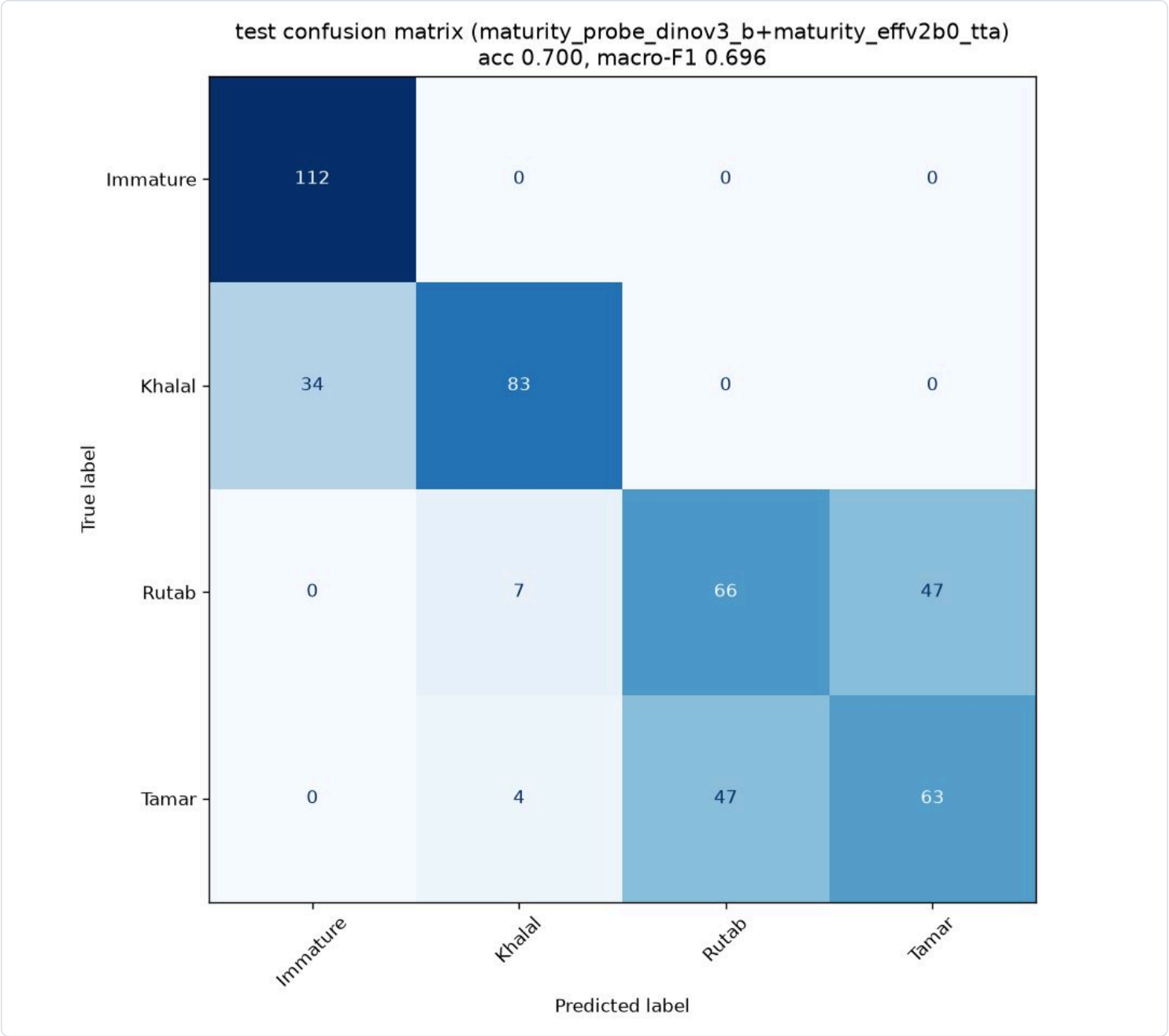
Per-class results (test)

Class	Precision	Recall	F1	n
Immature	0.767	1.000	0.868	112
Khalal	0.883	0.709	0.787	117
Rutab	0.584	0.550	0.567	120
Tamar	0.573	0.553	0.562	114

Accuracy by source

Source / variety	Accuracy
alhamdan	100.0%
morocco-kholt	66.7%

Confusion matrix (test)



Calibration does not transfer to the unseen test variety. Temperature scaling fitted on validation (seen varieties) makes test calibration worse, and the review threshold does not reach its target: on a new variety the model is confidently wrong, so its confidence cannot be used to route fruits to review.

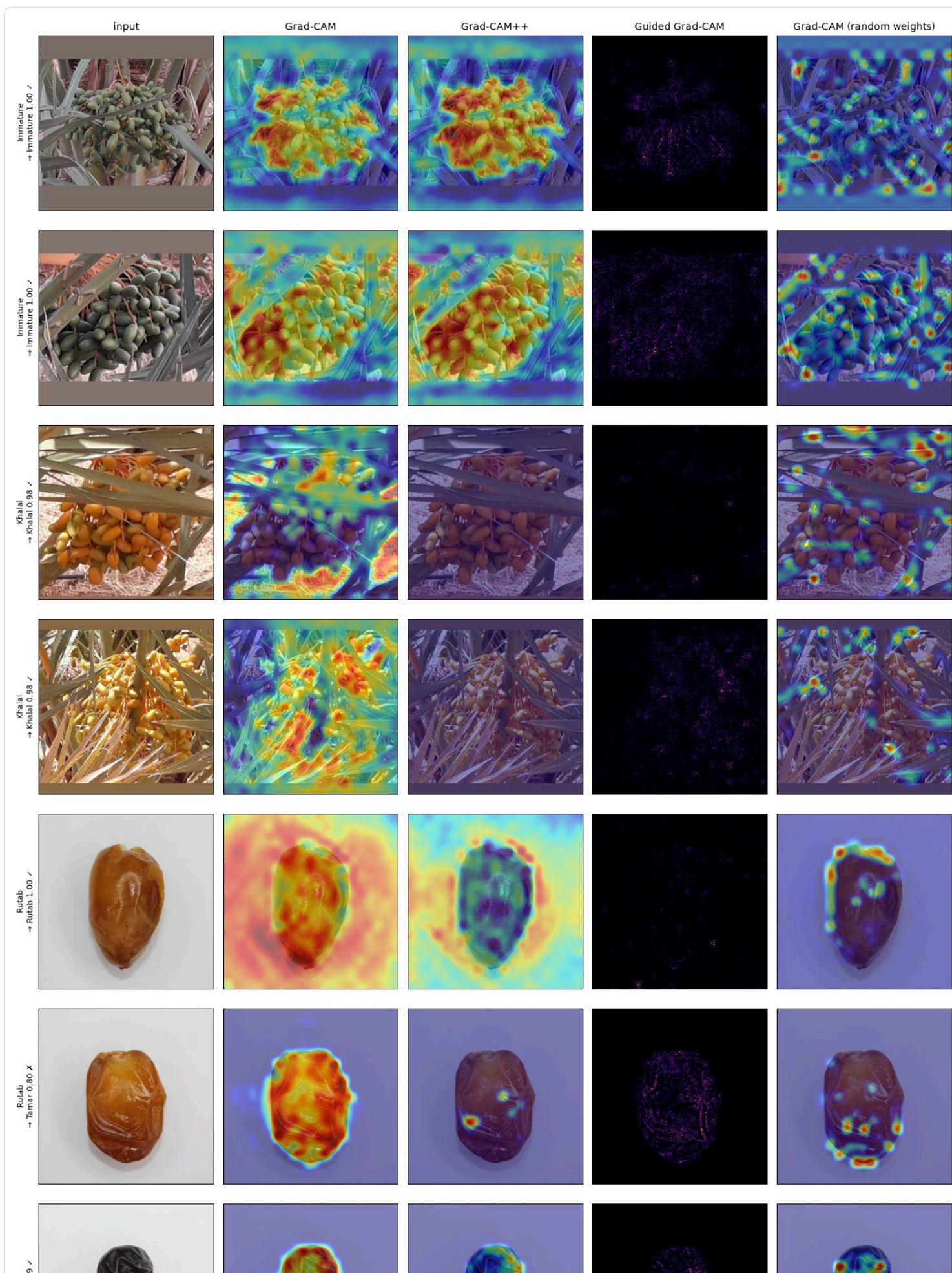
Calibration and manual-review threshold

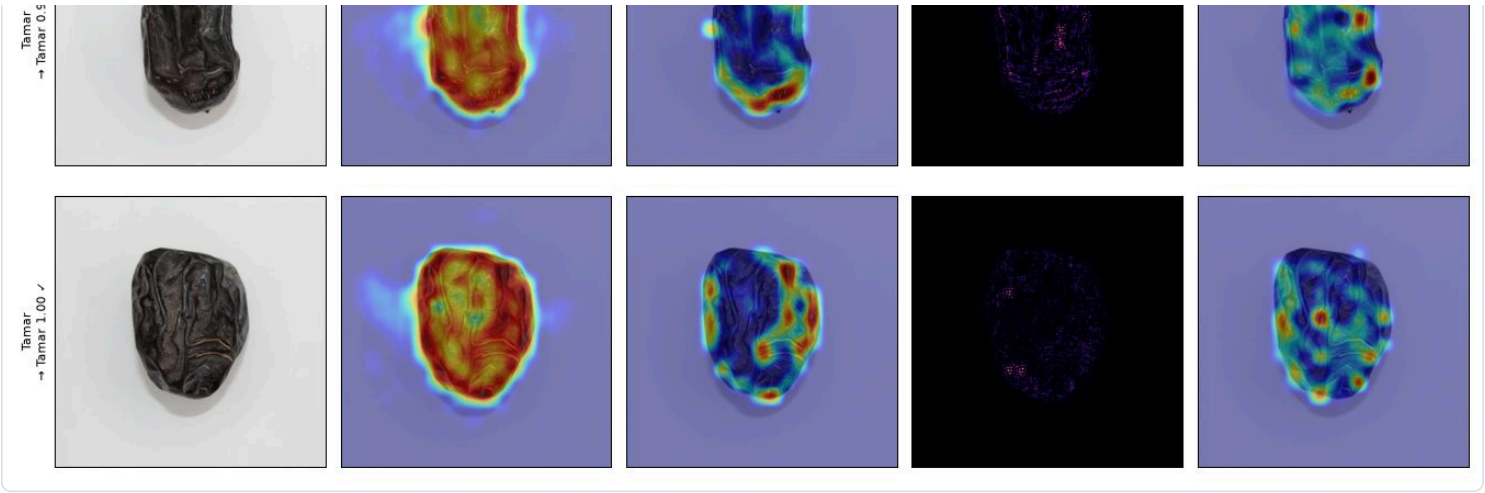
Temperature scaling (T = 0.5133) fitted on validation. Expected calibration error on test 0.115 → **0.208**; test NLL 0.727 → 1.075. The review threshold (0.937) is the lowest confidence whose *validation* accuracy reaches 99%; on test it auto-accepts **67.0%** of images at **81.6%** accuracy and sends 153 of 463 to manual review.

Confidence ≥	Auto-accepted	Accuracy on accepted
0.00	100.0%	70.0%
0.50	98.3%	71.2%
0.60	94.0%	72.9%
0.70	88.3%	74.8%
0.80	79.3%	77.7%
0.85	74.9%	79.0%
0.90	70.4%	79.8%
0.95	65.7%	82.2%
0.98	59.0%	86.5%

Explainability: Grad-CAM sanity checks (test split) · ensemble member DINOv3 ViT-B/16 336 probe

Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	0.094	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.163 vs 0.094	pass
Share of heatmap energy on fruit pixels vs fruit share of image	53.9% vs 43.3%	pass
Test accuracy with the background removed vs as photographed (single view)	65.2% vs 69.8%	pass

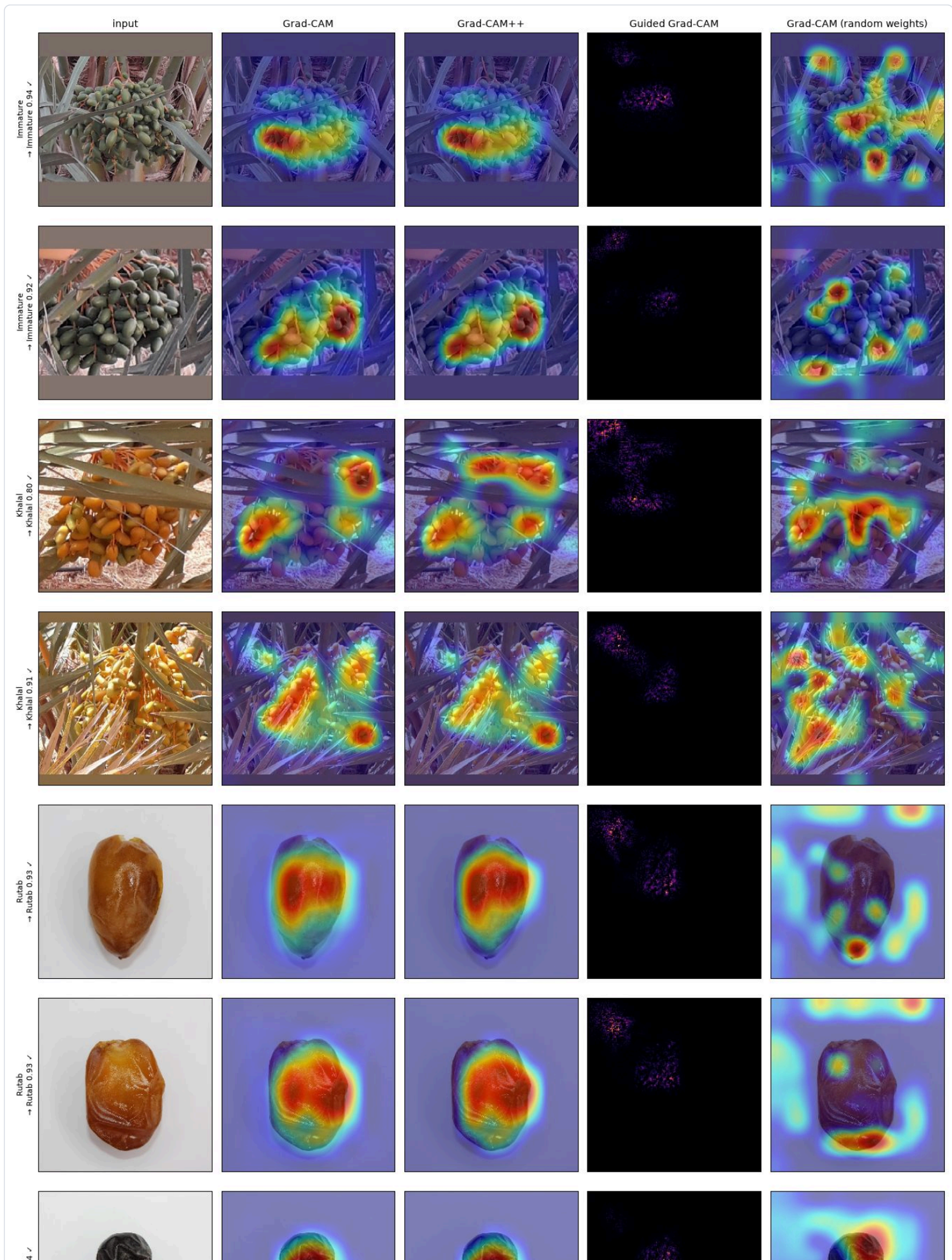


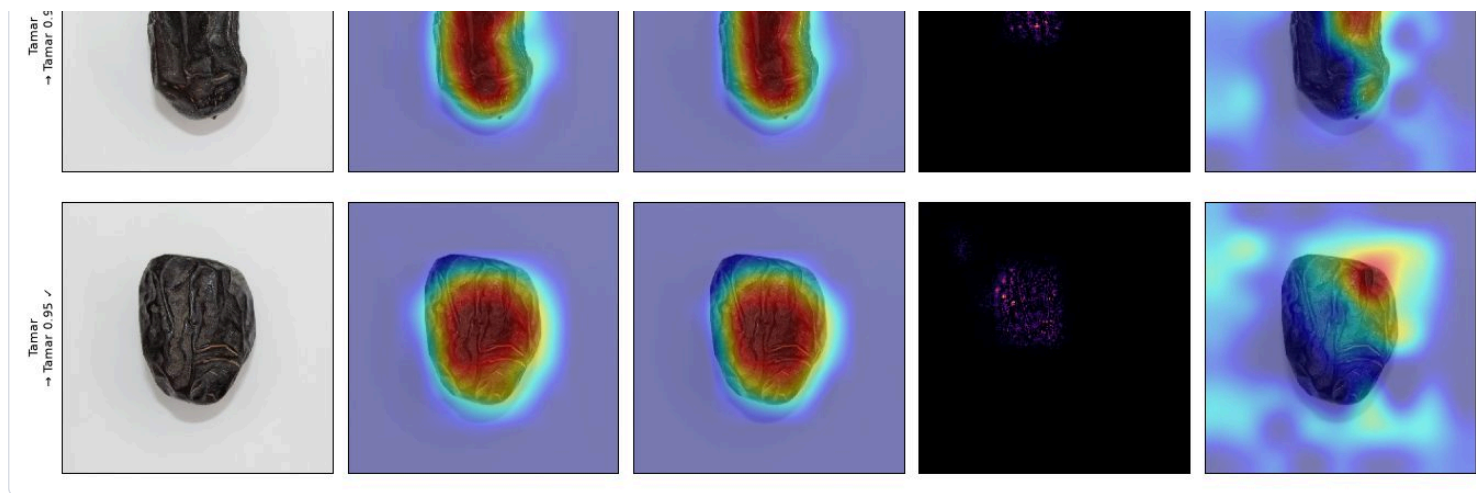


Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model. For the ViT, Grad-CAM runs on the final patch tokens (explain_vit.py).

Explainability: Grad-CAM sanity checks (test split) · ensemble member EfficientNetV2-B0 260

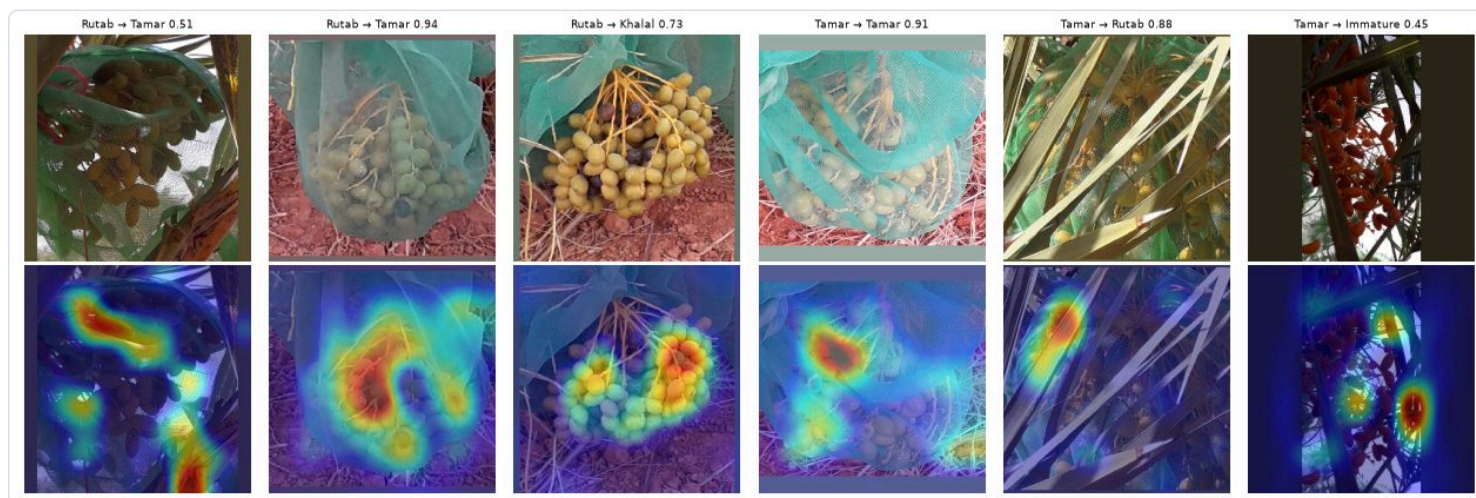
Check	Result	Verdict
Heatmap vs fully randomised model (Spearman, low = map depends on learning)	0.135	pass
Confidence drop when top-20% heatmap pixels are blurred vs random 20%	0.204 vs 0.079	pass
Share of heatmap energy on fruit pixels vs fruit share of image	76.0% vs 39.9%	pass





Columns: input · Grad-CAM · Grad-CAM++ · Guided Grad-CAM (Grad-CAM × SmoothGrad) · Grad-CAM of a randomised model.

Does the model look at the protective bags?



Grad-CAM on bagged Rutab/Tamar bunches of the unseen test variety: the heatmaps fall on the fruit visible through the netting, not on the net, so the bags are not being used as a shortcut. Most of these examples are misclassified, and the reason is visible: labels are per photo session, so a "Rutab" bunch still holds many green/yellow Khalal-coloured fruits, and a red "Tamar" bunch looks like another variety's earlier stage. Per-fruit stage labels on the dealer's own varieties are what would fix this.

TFLite export (float16)

Model	TFLite MB	Keras MB	Top-1 agreement with Keras	Max prob. diff	Median CPU latency
EfficientNetV2-B0 260	11.79	71.18	100.0% (n=200)	0.0039	67.8 ms

Phase 2 · Colour, defects, skin, weight (measurement rules)

No public dataset labels date defects, skin condition or weight, so these features are computed with transparent colour/texture rules on the segmented fruit rather than trained models. They cannot be scored with accuracy yet. As a sanity check, they are compared with the grade labels of the two varieties that have all three grades: defect area should rise from Grade-1 to Grade-3 (positive Spearman ρ with the grade number).

Feature	Gajar, Grade-1	Gajar, Grade-2	Gajar, Grade-3	Kupro, Grade-1	Kupro, Grade-2	Kupro, Grade-3	ρ Gajar	ρ Kupro
crack_pct	0.007	0.033	0.157	0.009	0.007	0.041	+0.31	+0.08
black_spot_pct	0.008	0.056	0.259	0.006	0.004	0.020	+0.48	+0.06
insect_hole_pct	0.011	0.043	0.186	0.003	0.003	0.011	+0.59	+0.15
sunburn_pct	0.000	0.000	0.000	0.014	0.000	0.000	–	-0.04
mold_pct	0.105	0.042	0.068	1.586	1.084	0.368	-0.05	-0.21
total_defect_pct	0.132	0.175	0.670	1.618	1.099	0.441	+0.53	-0.18
sugar_spot_pct	1.101	1.147	1.190	0.966	1.207	1.665	+0.09	+0.28
wrinkle_index	0.076	0.075	0.061	0.049	0.058	0.069	-0.25	+0.35
gloss_index	0.129	0.058	0.024	0.511	0.342	0.147	-0.31	-0.23
color_uniformity	0.403	0.402	0.402	0.384	0.382	0.373	-0.01	-0.08

What each feature measures

Feature	Method	Validation status
Colour uniformity	Mean CIELAB ΔE of skin pixels from their mean $\rightarrow$ score $\exp(-\Delta E/15)$; share within ΔE 10; dominant colours (k-means)	Deterministic measurement
Cracks / black spots / insect holes	Pixels ≥ 12 L* darker than their local neighbourhood; blobs split by elongation and circularity	ρ with grade +0.31 / +0.48 / +0.59 on Gajar; weak on Kupro
Sunburn / mold	Light, desaturated regions away from the rim and specular highlights; mold = compact, textured or greenish	Not validated; mold threshold raised to 2 % after false positives on glossy Kupro
Wrinkling	Edge density inside the fruit	Rises with grade on Kupro (+0.35), not Gajar (–0.25)
Gloss	Share and contrast of highlights relative to the fruit's own brightness	Falls with grade on both (–0.31, –0.23)
Sugar spots	Small light, desaturated granular blobs	Weak (+0.09 / +0.28)
Maturity by colour	Green share $\rightarrow$ Immature; yellow/red share $\rightarrow$ Khalal / partly ripened Rutab	Cannot split Rutab from Tamar (38 % on Alhamdan), so it only complements the CNN
Weight	Ellipsoid of revolution: $k \cdot L \cdot W^2$ with $k = 6.8 \times 10^{-4}$ g/mm ³ ($\rho \approx 1.3$ g/cm ³); needs an ArUco marker or mm/px	Untested on weighed fruit. A bag's scale reading recalibrates k per batch (--bag-weight)

Production readiness

Item	Status
Inference API	<code>grade.load_models()</code> + <code>grade.grade_image(path)</code> → JSON report; CLI <code>python src/grade.py img1 img2 ...</code>
Automated tests	16 passed in 30.78s (pytest tests/, 2026-09-29)
Input validation	Missing/unreadable/tiny images, images without fruit and invalid scale or bag weight raise <code>ValueError</code> ; the CLI reports each failure and exits non-zero
Confidence handling	Temperature-calibrated probabilities; predictions below the review threshold are flagged <code>needs_review</code> per fruit
Multiple fruits	Touching fruits are split at their neck (watershed); 451/451 grading test images counted as exactly one fruit
Pinned environment	<code>requirements-lock.txt</code> (Python 3.12, TensorFlow 2.18.1 + tensorflow-metal)
Edge deployment	TFLite float16 export of the Keras models with a parity check (<code>scripts/export_tflite.py</code>). The DINOv3 probe members run in PyTorch (Apple GPU or CPU); they have no mobile export yet
Reproducibility	Fixed seeds, split files saved (<code>data/splits.csv</code> , <code>data/quality/*_index.csv</code>), every number in this report regenerated from metric files

Limitations and what is needed before field use

1. **No dealer photos yet.** All scores are on public datasets. Eight of the requested varieties (Barhi, Mabroum, Khasab, Khasab Khalal, Helwa, Helwa Macnooz, Maktoumi, Bowythra, Kelas, Rashudia) are not in any public dataset. Accuracy on the dealer's own camera setup must be measured on a labelled sample of their photos before relying on it.
2. **Maturity data is orchard bunches.** Immature/Khalal/Rutab/Tamar labels are per bunch photo; many Rutab/Tamar bunches are inside protective bags. No public single-fruit Khalal images exist.
3. **Defects, skin and weight are unvalidated rules.** They need a few hundred dealer photos with marked defects, and 20–30 fruits weighed on a scale and photographed with the marker.
4. **Grade and size labels are subjective.** Most grade errors are between adjacent grades (1↔2); Gajar size classes overlap in measured size.
5. **99 % accuracy** is reached only on auto-accepted predictions (see the review thresholds); the remaining images go to a person. Headline accuracies above are over all test images.